MGM University

Chhatrapati Sambhajinagar

MGMUNIVERSITY

Name of Faculty – Engineering & Technology

Name of College – School of Engineering & Technology

Name of Department – Computer Science and Engineering

Name of Programme – B. Tech. Computer Science and Engineering CSE (Cybersecurity)

CURRICULUM BOOKLET

(With effect from Academic year 2025-26)

MGM University

Chhatrapati Sambhajinagar

MGMUNIVERSITY

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MGM University

Vision

- To ensure sustainable human development which encourages self-reliant and self-content society.
- To promote activities related to community services, social welfare and also Indian heritage and culture.
- To inculcate the culture of non-violence and truthfulness through vipassanna meditation and Gandhian Philosophy.
- To develop the culture of simple living and high thinking

Mission

- To impart state of art education and technical expertise to students and give necessary training to teachers to create self-reliant society for future.
- To encourage students to participate in Indian and International activities in sports, literature, etc. so that future generation becomes base for free and liberal society
- To educate students in areas like Management, Finance, Human relations to inculcate philosophy of simple living and high thinking value of simple economic society.
- To inculcate culture of non-violence and truthfulness through Vipassana.

  To sustain activities of Indian culture (viz. classical dance, music and fine arts) through establishing institutes like Mahagami, Naturopathy, etc.

विद्यापीठ गीत

अत्त दिप भव भव प्रदिप भव,

स्वरूप रूप भव हो

ज्ञान सब्ब विज्ञान सब्ब भव ,

सब्ब दिप भव हो

अत्ताहि अत्त नो नाथो ,

अत्ताहि अत्त नो गति

अत्त मार्गपर अप्रमादसे है तुझे चलना

सब्ब का कल्याण हो ,

वो कार्यकुशल करना

सब्ब का उत्तम मंगल , पथप्रदर्शक हो

अत्त दिप भव भव प्रदिप भव ,

स्वरूप रूप भव हो

ज्ञान सब्ब विज्ञान सब्ब भव ,

सब्ब दिप भव हो

बुद्धमं शरनं गच्छामि :

धम्मं शरनं गच्छामि :

संघं शरनं गच्छामि :

School of Engineering and Technology (SOET) at a Glance

The School of Engineering and Technology (SoET), a self-aided, non-minority, conducted college of MGM University is the first institution to provide integrated B.Tech and M.Tech programs in the Marathwada region of Maharashtra. A sister institution of the illustrious Jawaharlal Nehru Engineering College (JNEC), Aurangabad, it was established as MGM Polytechnic College in 2010. However, recognizing the rising need of quality engineering education after schooling in the region and beyond, it was transformed into the School of Engineering and Technology (SOET) in 2022.

Inheriting a 39 years rich legacy of imparting world-class professional education from MGM Trust, SoET provides ultramodern research and innovation facilities to its students. Its integrated programs eliminate the need to appear for 12th standard and various engineering entrance tests, thus saving valuable time and energy of the students. It allows the students to utilize the same time to gain more practical knowledge of the field and a strong edge over the competition.

The college offers a wide variety of Undergraduate (Including Six Years Integrated Programs after 10th), Postgraduate (Including Five Years Integrated Programs after 12th) and PG Diploma programs to fulfil the dreams of the students. All these are advanced engineering programs founded on industry-oriented curriculum.

The Institution employs many modern teaching-learning methods like Project-based Learning, Research-based Learning and Collaborative Learning to develop comprehensive understanding and real-life skills in the students. The School has also signed several memoranda of understanding with renowned institutions, associations and industries to provide internship and project work opportunities to the students. It also joined hands with COURSERA, NPTEL, SWAYAM, ATAL and other MOOC platforms to facilitate blended learning facilities for the students.

Unique in its structure, methods and goals, the institution is strongly rooted in the philosophy of training and research that enhances the relationship between knowledge and its application and seeks to promote the creation of an ideal society through holistic education.

Vision

Enrich the students in technical education to work globally with human values.

Mission

- To inculcate technical skills by continuous assessment practices.
- To strengthen interaction between industry and institute.
- To involve the students in social activities with environmental awareness.

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Programs offered at SOET

Undergraduate Programmes	Postgraduate Programmes	PhDProgrammes	PG DiplomaProgrammes	CertificateProgrammes
B.Tech (Integrated) Civil Engineering with Computer Applications	M. Tech (Integrated) Computer Science and Engineering (AI and Machine Learning)	Ph.D. Energy Engineering	**PG Diploma**In Construction Management	Diploma In Electric Vehicle Technology
B.Tech (Integrated) Computer Science and Engineering (Data Science)	M. Tech Energy Engineering	Ph.D. Medical Technology		Diploma In Drone Technology
B.Tech (Integrated) Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)	M. Tech Medical Technology	Ph.D. Industrial Engineering and Management		
B.Tech (Integrated) Electronics and Computer Engineering	M. Tech Industrial Engineering and Management	Ph.D. Design Engineering		
B.Tech (Integrated) Industrial Engineering	M. Tech Design Engineering			
B.Tech (Integrated) Advanced Mechatronics and Industrial Automation				

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B.Tech (Integrated) Robotics and AI				
B.Tech (Integrated) Electrical and Computer Engineering				
B.Tech (Integrated) Information Technology				
B.Tech Agriculture Engineering				
B.Tech Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)				
B.Tech Electrical and Instrumentation Engineering				
B.Tech Electronics Engineering (VLSI Design and Technology)				
B.Tech Pharmaceutical Engineering				
B.Tech Computer Science and Design				

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Department of Computer Science and Engineering

The Bachelor of Technology (B. Tech.) in Computer Science and Engineering degree program started in the year 1983 and is approved by the All India Council for Technical Education (AICTE), New Delhi and offers Choice Based Credit System education. In addition to core courses, students can opt for discipline specific elective subjects, open elective subjects from different institutes of the University. In addition, this program is uniquely designed to increase the employability and to prepare students to work in a Multi-disciplinary work environment. The program offers Major degrees in Computer Science and Engineering and is open to students opting for minor specializations as per their interests. Pedagogies concentrating on student's active participation are extensively used in the teaching-learning process.

Vision

To provide update technology in computer engineering to fulfill needs of industry and society.

Mission

- To impart technical knowledge and skills by involving students in educational activities.
- To enable students to be entrepreneurs enhancing interaction between institute and industry.
- To nurture moral and professional values to build up society and nation.

Eligibility –

1. Maharashtra State Candidate.

(i) The Candidate should be an Indian National and having domicile of Maharashtra state and/or born in Maharashtra state.

(ii) Passed HSC or its equivalent examination with Physics and Mathematics as compulsory subjects along with one of the Chemistry or Biotechnology or Biology or Technical Vocational subject or Computer Science or Information Technology or Informatics Practices or Agriculture or Engineering Graphics or Business Studies, and obtained at least 45% marks (at least 40% marks, in case of Backward class categories and Persons with Disability candidates belonging to Maharashtra State only) in the above subjects taken together and the candidate should have appeared in MGMU-CET 2022/ MHT-CET 2022/ PERA CET 2022/ JEE (Main) Paper-I 2022 and should obtain non zero score in MGMU-CET 2022/ MHT-CET 2022/ PERA CET 2022/ JEE (Main) Paper-I 2022. However, preference shall be given to the candidate obtaining non-zero positive score in MGMU-CET 2022 over the candidates who obtained non-zero score in MHT-CET 2022/ PERA CET 2022.

OR

(ii)Passed Diploma in Engineering and Technology and obtained at least 45% marks (at least 40% marks, in case of Backward class categories and Persons with Disability candidates belonging to Maharashtra State only).

2. All India Candidates –

(i) The Candidate should be an Indian National.

(ii) Passed HSC or its equivalent examination with Physics and Mathematics as compulsory subjects along with one of the Chemistry or Biotechnology or Biology or Technical Vocational subject or Computer Science or Information Technology or Informatics Practices or Agriculture or Engineering Graphics or Business Studies , and obtained at least 45% marks (at least 40% marks, in case of Backward class categories and Persons with Disability candidates belonging to Maharashtra State only) in the above subjects taken together and candidate should have appeared in MGMU-CET 2022/ MHT-CET 2022/ PERA CET 2022/ JEE (Main) Paper-I 2022 and should

obtain non-zero score in MGMU-CET 2022/ MHT-CET 2022/ PERA CET 2022/ JEE (Main) Paper-I 2022. However, preference shall be given to the candidate obtaining non-zero positive score in JEE Mains Paper-I over the candidates who obtained non-zero score in MGMU-CET 2022/ MHT-CET 2022/ PERA CET 2022.

OR

(ii) Passed Diploma in Engineering and Technology and obtained at least 45% marks (at least 40% marks, in case of Backward class categories and Persons with Disability candidates belonging to Maharashtra State only)

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Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Computer Science and Engineering (Cybersecurity)

Programme Type (UG/PG): UG **Duration:** 04 YEAR

First Year - Semester I (Group A)												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS21BSL101	Single and Multivariable Calculus	Theory	4	4	-	60	40	100		16	40
BSC	APS21BSL102	Engineering Physics	Theory	3	3	-	60	40	100		16	40
ESC	APS21ESL101	Python Programming	Theory	2	2	-	60	40	100		16	40
ESC	APS21ESL102	Engineering Graphics	Theory	2	2	-	60	40	100		16	40
AEC	MGM54AEL101	Communicative English	Theory	1	1	-	30	20	50		8	20
VSEC	APS21VSP101	Engineering Exploration	Practical	2	-	4	60	40	100		16	40
BSC	APS21BSP101	Engineering Physics Lab	Practical	1	-	2	30	20	50		8	20
ESC	APS21ESP101	Python Programming Lab	Practical	1	-	2	30	20	50		8	20
ESC	APS21ESP102	Engineering Graphics Studio	Practical	2	-	4	30	20	50		8	20
ESC	APS21ESP103	Recent Trends in Integrated Technology	Practical	1	-	2	30	20	50		8	20
AEC	MGM54AEP101	Communicative English Lab	Practical	1	-	2	30	20	50		8	20
CCA	MGM82CCP101 MGM82CCP102 MGM82CCP103	NCC / Yoga / Sports	Practical	2	-	4	30	20	50		8	20
Total				22	12	20	510	340	850	0	136	340

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation,

Course Category: BSC-Basic Science Course, ESC-Engineering Science Course, VSEC-Vocational Skill and Skill Enhancement Courses, PCC-Programme Core Course, PEC-Programme Elective Course, MDM-Multidisciplinary Minor, HSSM-Entrepreneurship/Economics/Management Courses, Lib-Liberal Learning Course, MIN-Minor, OE-Generic / Open electives, VSC-Vocational skill course, SEC-Skill Enhancement course, AEC-Ability Enhancement course, IKS-Indian Knowledge system, VEC-Value Education course, OJT-On Job Training / Internship / Apprenticeship, FP-Field project, CEP-Community engagement and service, CC-Co – curricular course, RM-Research methodology, RP-Research project.

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)

Programme Type (UG/PG): UG **Duration:** 04 YEAR

First Year- Semester II (Group A)

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact hrs/ week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
BSC	APS21BSL103	Linear Algebra and Differential Equations	Theory	4	4	-	60	40	100	-	16	40
BSC	APS21BSL104	Engineering Chemistry	Theory	3	3	-	60	40	100	-	16	40
ESC	APS21ESL103	Engineering Mechanics	Theory	2	2	-	60	40	100	-	16	40
ESC	APS21ESL104	Building Programming logic in C	Theory	1	1	-	30	20	50	-	8	20
PCC	APS21PCL101	Basics of Electrical and Electronics Engineering	Theory	2	2	-	60	40	100	-	16	40
IKS	APS21IKL1XX	Indian Knowledge System	Theory	2	2	-	60	40	100	-	16	40
VSEC	APS21VSP102	Workshop Practices	Practical	2	-	4	60	40	100	-	16	40
BSC	APS21BSP102	Engineering Chemistry Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP104	Engineering Mechanics Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP105	Building Programming logic in C Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	APS21PCP101	Electrical and Electronics Technology Lab	Practical	1	-	2	30	20	50	-	8	20
CCA	MGM82CCP104 MGM73CCP105 MGM73CCP106	NSS/ Fine Art/ Visual Art	Practical	2	-	4	30	20	50	-	8	20
Total				22	14	16	540	360	900	0	144	360

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation,

Course Category: BSC-Basic Science Course, ESC-Engineering Science Course, VSEC-Vocational Skill and Skill Enhancement Courses, PCC-Programme Core Course, PEC-Programme Elective Course, MDM-Multidisciplinary Minor, HSSM-Entrepreneurship/Economics/Management Courses, Lib-Liberal Learning Course, MIN-Minor, OE-Generic / Open electives, VSC-Vocational skill course, SEC-Skill Enhancement course, AEC-Ability Enhancement course, IKS-Indian Knowledge system, VEC-Value Education course, OJT-On Job Training / Internship / Apprenticeship, FP-Field project, CEP-Community engagement and service, CC-Co – curricular course, RM-Research methodology, RP-Research project.

Level 4.5 Award of UG certificate with 40 credits and an additional 4-credits core NSQF course / internship OR continue with major and minor

First Year - Semester I (Group B)												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS21BSL101	Single and Multivariable Calculus	Theory	4	4	-	60	40	100	-	16	40
BSC	APS21BSL104	Engineering CHEMISTRY	Theory	3	3	-	60	40	100	-	16	40
ESC	APS21ESL101	Python Programming	Theory	2	2	-	60	40	100	-	16	40
ESC	APS21ESL103	Engineering Mechanics	Theory	2	2	-	60	40	100	-	16	40
AEC	MGM54AEL101	Communicative English	Theory	1	1	-	30	20	50	-	8	20
PCC	APS21PCL101	Basics of Electrical and Electronics Engineering	Theory	2	2	-	60	40	100	-	16	40
VSEC	APS21VSP102	Workshop Practices	Practical	2	-	4	60	40	100	-	16	40
BSC	APS21BSP102	Engineering Chemistry Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP101	Python Programming Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS21ESP104	Engineering Mechanics Lab	Practical	1	-	2	30	20	50	-	8	20
AEC	MGM54AEP101	Communicative English Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	APS21PCP101	Electrical and Electronics Technology Lab	Practical	1	-	2	30	20	50	-	8	20
CCA	MGM82CCP101 MGM82CCP102 MGM82CCP103	NCC / Yoga / Sports	Practical	2	-	4	30	20	50	-	8	20
Total				23	14	18	570	380	950	0	88	64

First Year - Semester II (Group B)												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS21BSL103	Linear Algebra and Differential Equations	Theory	4	4	-	60	40	100	-	16	40
BSC	APS21BS	Engineering	Theo	3	3	-	60	40	100	-	16	40

	L102	Physics	ry									
ESC	APS21ESL102	Engineering Graphics	Theory	2	2	-	60	40	100	-	16	40
IKS	APS21IKL1XX	Indian Knowledge System	Theory	2	2	-	60	40	100	-	16	40
ESC	APS21ESL104	Building Programming logic in C	Theory	1	1	-	30	20	50	-	8	20
VSEC	APS21VSP101	Engineering Exploration	Practical	2	-	4	60	40	100	-	16	40
BSC	APS21BSP101	Engineering Physics Lab	Practical	1	-	2	30	20	50		8	20
ESC	APS21ESP102	Engineering Graphics Studio	Practical	2	-	4	30	20	50		8	20
ESC	APS21ESP105	Building Programming logic in C Lab	Practical	1	-	2	30	20	50		8	20
ESC	APS21ESP103	Recent Trends in Integrated Technology	Practical	1	-	2	30	20	50		8	20
CCA	MGM82CCP104 MGM73CCP105 MGM73CCP106	NSS/ Fine Art/ Visual Art	Practical	2	-	4	30	20	50		8	20
Total				21	12	18	480	320	800	0	128	320

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Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology
Name of the Programme: B.Tech Computer Science and Engineering (Cybersecurity)
Programme Type (UG/PG): UG **Duration:** 04 YEAR

Semester III												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	SCS24PCL201	Discrete Mathematics	Lecture	3	3	-	60	40	100	-	16	40
PCC	SCS24PCL202	Data Structure	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL203	CodeLock: Cybersecurity & Cryptography Essentials	Lecture	3	3	-	60	40	100	-	16	40
EEMC	SCS24HSL204	Business Management & Financial Accounting	Lecture	2	2	-	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50	-	8	20
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50	-	8	20
VEC	MGM56VEL102	Constitution of India	Lecture	2	2	-	30	20	50	-	8	20
PCC	SCS24PCP202	Data Structure Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	SCS24PCP203	CodeLock: Cybersecurity & Cryptography Essentials Lab	Practical	1	-	2	30	20	50	-	8	20
FP	SCS24FPJ205	Web Technology Lab	Practical	2		4	50	-	50	20	-	20
		TOTAL		22	18	8	500	300	800			

Note:
Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation,
Course Category: BSC-Basic Science Course, ESC-Engineering Science Course, VSEC-Vocational Skill and Skill Enhancement Courses, PCC-Programme Core Course, PEC-Programme Elective Course, MDM-Multidisciplinary Minor, HSSM-Entrepreneurship/Economics/Management Courses, Lib-Liberal Learning Course, MIN-Minor, OE-Generic / Open electives, VSC-Vocational skill course, SEC-Skill Enhancement course, AEC-Ability Enhancement course, IKS-Indian Knowledge system, VEC-Value Education course, OJT-On Job Training / Internship / Apprenticeship, FP-Field project, CEP-Community engagement and service, CC-Co – curricular course, RM-Research methodology, RP-Research project.

For Honor course (**Please refer the Annexure B.**)

For Specialization Minor course (Please refer the Annexure C.)

Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology
Name of the Programme: B.Tech Computer Science and Engineering (Cybersecurity)
Programme Type (UG/PG): UG **Duration:** 04 YEAR

Semester IV												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	SCS24PCL251	Ethical Hacking	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL252	Operating System	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL253	Object Oriented Programming Using C++	Lecture	2	2	-	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50	-	8	20
EEMC	SCS24HSL254	Enterprenership Development	Lecture	2	2	-	60	40	100	-	16	40
AEC		Refer MGMU AEC Basket	Lecture	2	2	-	30	20	50	-	8	20
VEC	MGM21VEL101	Environmental Studies	Lecture	2	2	-	30	20	50	-	8	20
PCC	SCS24VSL255	Blockchain	Lecture	1	1	-	30	20	50	-	8	20
PCC	SCS24PCP251	Ethical Hacking Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	SCS24PCP252	Operating System Lab	Practical	1	-	2	30	20	50	-	8	20
VSEC	SCS24PCP253	Object Oriented Programming Using C++Lab	Practical	1	-	2	30	20	50	-	8	20
	SCS24VSP255	Blockchain Lab	Practical	1	-	2	30	20	50	-	8	20

Note:
Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation,
Course Category: BSC-Basic Science Course, ESC-Engineering Science Course, VSEC-Vocational Skill and Skill Enhancement Courses, PCC-Programme Core Course, PEC-Programme Elective Course, MDM-Multidisciplinary Minor, HSSM-Entrepreneurship/Economics/Management Courses, Lib-Liberal Learning Course, MIN-Minor, OE-Generic / Open electives, VSC-Vocational skill course, SEC-Skill Enhancement course, AEC-Ability Enhancement course, IKS-Indian Knowledge system, VEC-Value Education course, OJT-On Job Training / Internship / Apprenticeship, FP-Field project, CEP-Community engagement and service, CC-Co – curricular course, RM-Research methodology, RP-Research project.

•Exit option to qualify for UG in Computer Science and Engineering: After securing all credits of first and second year and provided the student secures additional 8 credits in skill-based vocational courses (skill-based courses, internship, mini projects etc) offered during summer vacation after the first year or second year .(As per Annexture - I)

• Skill-based vocational course : 08 credits

For Honor course (**Please refer the Annexure B.**)

For Specialization Minor course (Please refer the Annexure C.)

MGMUNIVERSITY

Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology
Name of the Programme: B.Tech Computer Science and Engineering (Cybersecurity)
Programme Type (UG/PG): UG **Duration:** 04 YEAR

Semester V												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	SCS24PCL301	Computer Network	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL302	Cyber Law & Policies	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL303	Data Base Management System	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL304	Advanced Cryptography	Lecture	2	2	-	60	40	100	-	16	40
PEC-1.1	SCS24PEL305/ SCS24PEL306	Basics of Cloud Computing/ Introduction to DevOps	Lecture	3	3	-	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	4	4	-	60	40	100	-	16	40
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50	-	8	20
PCC	SCS24PCP301	Computer Network Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	SCS24PCP307	Web Application Security Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	SCS24PCP303	Data Base Management System Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	SCS24PCP304	Advanced Cryptography Lab	Practical	1	-	2	30	20	50	-	8	20
PEC-1.1	SCS24PEP305/ SCS24PEP306	Basics of Cloud Computing Lab / Introduction to DevOps Lab	Practical	1	-	2	30	20	50	-	8	20
		TOTAL		**22**	**17**	**10**	**540**	**360**	**900**			

Note:
Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation,
Course Category: BSC-Basic Science Course, ESC-Engineering Science Course, VSEC-Vocational Skill and Skill Enhancement Courses, PCC-Programme Core Course, PEC-Programme Elective Course, MDM-Multidisciplinary Minor, HSSM-Entrepreneurship/Economics/Management Courses, Lib-Liberal Learning Course, MIN-Minor, OE-Generic / Open electives, VSC-Vocational skill course, SEC-Skill Enhancement course, AEC-Ability Enhancement course, IKS-Indian Knowledge system, VEC-Value Education course, OJT-On Job Training / Internship / Apprenticeship, FP-Field project, CEP-Community engagement and service, CC-Co – curricular course, RM-Research methodology, RP-Research project.

For Honor course (**Please refer the Annexure B.**)

For Specialization Minor course (Please refer the Annexure C.)

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Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology
Name of the Programme: B.Tech Computer Science and Engineering (Cybersecurity)
Programme Type (UG/PG): UG **Duration:** 04 YEAR

Semester VI												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	SCS24PCL351	Cloud Security	Lecture	3	3	-	60	40	100	-	16	40
PCC	SCS24PCL352	Machine Learning for Cybersecurity	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL353	Blockchain Security	Lecture	2	2	-	60	40	100	-	16	40
PEC-1.2	SCS24PEL354/ SCS24PEL355	Cloud Computing using AWS/ DevOps Zenith (Advance DevOps)	Lecture	3	3	-	60	40	100	-	16	40
PEC-2.1	SCS24PEL356/ SCS24PEL357	Image Processing/ Quantum Computing	Lecture	3	3	-	60	40	100	-	16	40
VSEC	SCS24VSL358	Software Testing	Lecture	1	1	-	30	20	50	-	8	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCP351	Cloud Security Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	SCS24PCP352	Machine Learning for Cybersecurity Lab	Practical	1	-	2	30	20	50	-	8	20
PEC-1.2	SCS24PEP354/ SCS24PEP355	Cloud Computing using AWS Lab / DevOps Zenith (Advance DevOps) Lab	Practical	1	-	2	30	20	50	-	8	20
PEC-2.1	SCS24PEP356/ SCS24PEP357	Image Processing Lab/ Quantum Computing Lab	Practical	1	-	2	30	20	50	-	8	20
VSEC	SCS24VSP358	Software Testing Lab	Practical	1	-	2	30	20	50	-	8	20
		TOTAL		**21**	**16**	**10**	**540**	**360**	**900**			

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation,
Course Category: BSC-Basic Science Course, ESC-Engineering Science Course, VSEC-Vocational Skill and Skill Enhancement Courses, PCC-Programme Core Course, PEC-Programme Elective Course, MDM-Multidisciplinary Minor, HSSM-Entrepreneurship/Economics/Management Courses, Lib-Liberal Learning Course, MIN-Minor, OE-Generic / Open electives, VSC-Vocational skill course, SEC-Skill Enhancement course, AEC-Ability Enhancement course, IKS-Indian Knowledge system, VEC-Value Education course, OJT-On Job Training / Internship / Apprenticeship, FP-Field project, CEP-Community engagement and service, CC-Co – curricular course, RM-Research methodology, RP-Research project.

- Exit option to qualify for UG in Computer Science and Engineering: After securing all credits of first and second year and provided the student secures additional 8 credits in skill-based vocational courses (skill-based courses, internship, mini projects etc) offered during summer vacation after the first year or second year .(As per Annexture - I)
- Skill-based vocational course : 08 credits

For Honor course (**Please refer the Annexure B.**)

For Specialization Minor course (Please refer the Annexure C.)

Students who are eligible (having CGPA >= 7.5 at Third year) can opt for Honors with Research degree, where the students will have to work on a research project or dissertation for 18 credits in the fourth year in the same engineering / Technology discipline, distributed over semester 7th and 8th . List of Honors with Research Courses is given in Annexure D.

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Computer Science and Engineering (Cybersecurity)

Programme Type (UG/PG): UG **Duration:** 04 YEAR

Semester VII													
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	SCS24PCL401	Emerging Threats in Cybersecurity	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL402	Cyber Risk Management & Compliance	Lecture	2	2	-	60	40	100	-	16	40
PEC	SCS24PEL403	Big Data/Data Analytics	Lecture	2	2	-	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
Internship	SCS24JTI404	Internship	Practical	12	-	24	60	90	150	-	20	60
		TOTAL		**20**	**8**	**24**	**300**	**250**	**550**			

Note:
Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation,
Course Category: BSC-Basic Science Course, ESC-Engineering Science Course, VSEC-Vocational Skill and Skill Enhancement Courses, PCC-Programme Core Course, PEC-Programme Elective Course, MDM-Multidisciplinary Minor, HSSM-Entrepreneurship/Economics/Management Courses, Lib-Liberal Learning Course, MIN-Minor, OE-Generic / Open electives, VSC-Vocational skill course, SEC-Skill Enhancement course, AEC-Ability Enhancement course, IKS-Indian Knowledge system, VEC-Value Education course, OJT-On Job Training / Internship / Apprenticeship, FP-Field project, CEP-Community engagement and service, CC-Co – curricular course, RM-Research methodology, RP-Research project.

For Honor course (**Please refer the Annexure B.**)

For Specialization Minor course (Please refer the Annexure C.)

For Honors with Research (Please refer the Annexure D.)

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech Computer Science and Engineering (Cybersecurity)

Programme Type (UG/PG): UG **Duration:** 04 YEAR

Semester VIII												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	SCS24PCL 451	Digital Forensics	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCL 452	Internet of Things (IoT) Security	Lecture	2	2	-	60	40	100	-	16	40
PEC-1.3	SCS24PEL 453/ SCS24PEL 454	Cloud Computing using Azure / DevOps- on AWS	Lecture	2	2	-	60	40	100	-	16	40
PEC-2.2	SCS24PEL 455/ SCS24PEL 456	Computer Vision/ AR VR	Lecture	2	2	-	60	40	100	-	16	40
RM	SCS24RML457	Research Methodology	Lecture	4	4	-	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
PCC	SCS24PCP 451	Digital Forensics Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	SCS24PCP 452	Internet of Things (IoT) Security Lab	Practical	1	-	2	30	20	50	-	8	20
PEC-1.3	SCS24PEP 453/ SCS24PEP 454	CC using Azure Lab / DevOps- on AWS Lab	Practical	1	-	2	30	20	50	-	8	20
PEC-2.2	SCS24PEP 455/ SCS24PEP 456	Computer Vision/ AR VR Lab	Practical	1	-	2	30	20	50	-	8	20
Project	SCS24RPJ 458	Major Project	Practical	4	-	8	60	40	100	-	16	40
		TOTAL		22	14	16	540	360	900			

Note:

Nature of Course: L- Lecture, P-Practical, S-Seminar, J-Project, I-Internship, D-Dissertation,

Course Category: BSC-Basic Science Course, ESC-Engineering Science Course, VSEC-Vocational Skill and Skill Enhancement Courses, PCC-Programme Core Course, PEC-Programme Elective Course, MDM-Multidisciplinary Minor, HSSM-Entrepreneurship/Economics/Management Courses, Lib-Liberal Learning Course, MIN-Minor, OE-Generic / Open electives, VSC-Vocational skill course, SEC-Skill Enhancement course, AEC-Ability Enhancement

course, IKS-Indian Knowledge system, VEC-Value Education course, OJT-On Job Training / Internship / Apprenticeship, FP-Field project, CEP-Community engagement and service, CC-Co – curricular course, RM-Research methodology, RP-Research project.

•Exit option to qualify for UG in Computer Science and Engineering: After securing all credits of first and second year and provided the student secures additional 8 credits in skill-based vocational courses (skill-based courses, internship, mini projects etc) offered during summer vacation after the first year or second year .(As per Annexture - I)

• Skill-based vocational course : 08 credits

For Honor course (**Please refer the Annexure B.**)

For Specialization Minor course (Please refer the Annexure C.)

For Honors with Research (Please refer the Annexure D.)

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Annexure –I

Exit Options after FY, SY and TY

1. Exit option to qualify for Certification, common at Institute level: After securing all credits of First Year and securing 8 credits in work-based vocational courses or internship / Apprenticeship offered during summer vacation will be awarded One Year UG certificate in Tech(Discipline) (As per Annexture- I).

   •Work-based vocational course: 08 credits

2. Exit option to qualify for UG Diploma in Computer Science and engineering With Multidisciplinary Minor: After securing all credits of first and second year and provided the student secures additional 8 credits in skill-based vocational courses (skill-based courses, internship, mini projects etc) offered during summer vacation after the first year or second year .(As per Annexure - I)

   • Skill-based vocational course: 08 credits

3. Exit option to qualify for B.Voc in Computer Science and engineering With Multidisciplinary Minor: After securing all credits of first, second and third year with additional 6 credits in skill-based vocational courses (skill-based courses, internship, mini projects etc) offered during summer vacation after the sixth semester. (As per Annexure - I).

NAME OF THE PROGRAMME: One Year UG Certificate in Tech. (Computer Science and engineering With Multidisciplinary Minor)

Name of the Work based Vocational courses (8credits)

Course Category	Course Code	Course Title	Nature of Course	No of Credits	Teaching (Contact hrs/ week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
VSC	SCS24VSI101	Computer Hardware and Maintainance	Practical	8	-	16	60	40	100	-	16	40

NAME OF THE PROGRAMME: UG Diploma in the Computer Science and engineering With Multidisciplinary Minor

Name of the skill-based vocational courses (8credits)

Course Category	Course Code	Course Title	Nature of Course	No of Credits	Teaching (Contact hrs/ week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
VSC	SCS24VSI201	Computer Network & Cyber Security	Practical	8	-	16	60	40	100	-	16	40

NAME OF THE PROGRAMME: B.Voc. in Computer Science and engineering With Multidisciplinary Minor

Name of the B.Voc. in the relevant Discipline (6credits)

Course Category	Course Code	Course Title	Nature of Course	No of Credits	Teaching (Contact hrs/ week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)		
					L	P	Internal	External	Total	Internal	External	Total
VSC	SCS24VSI301	Data Analytics	Practical	6	-	12	60	40	100	-	16	40

Annexure - B

Honor Offered

Sr. No.	Title of Honors	Program Offering
01	Data Science	B.Tech Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)
		B.Tech Computer Science and Design

Honor Structure

Students after FY can opt for B.Tech. Degree with Honours/ Double Minor. Eligibility for admission will be minimum CGPA of 7.5 at First Year.

For Honours: The students will have to take additional 5 courses of 20 credits in Computer Science and Engg discipline distributed over semesters III to VIII. (As per Annexure II)

NAME OF THE PROGRAMME: B.Tech Computer Science and Engineering (Cyber Security) with Multidisciplinary Minor Honor in (Data Science)

Course Category	Course Code	Course Title	Nature of Course	No of Credits	Teaching (Contact hrs/ week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)			Semester
					L	P	Internal	External	Total	Internal	External	Total	
PCC	CSB24HNL 201	Statistical Data Science	Lecture	3	3	-	60	40	100	-	16	40	III
PCC	CSB24HNP 201	Statistical Data Science Lab	Practical	1	-	2	30	20	50	-	8	20	
PCC	CSB24HNL 251	Python for Data Science	Lecture	3	3	-	60	40	100	-	16	40	IV
PCC	CSB24HNP 251	Python for Data Science Lab	Practical	1	-	2	30	20	50	-	8	20	
PCC	CSB24HNL 301	Machine Learning-1	Lecture	3	3	-	60	40	100	-	16	40	V
PCC	CSB24HNP 301	Machine Learning-1 Lab	Practical	1	-	2	30	20	50	-	8	20	
PCC	CSB24HNL 351	Machine Learning-2 & Big Data Analysis	Lecture	3	3	-	60	40	100	-	16	40	VI
PCC	CSB24HNP 351	Machine Learning-2 & Big Data Analysis Lab	Practical	1	-	2	30	20	50	-	8	20	
PCC	CSB24HNL 401	Case studies in Data Science & Project	Lecture	4	4	-	60	40	100	-	16	40	VIII
		TOTAL		**20**	**16**	**8**	**420**	**280**	**700**				

Annexure - D

Structure of Honors with Research

Additional Courses for MDM and Honors with Research

Students after TY can opt for the B.Tech. Degree Honors with Research. Eligibility for admission will be minimum CGPA of 7.5 at third year.

The students will work on a research project or dissertation for 18 credits in the IV year in the Computer Science and Engineering with Multidisciplinary Minor discipline distributed over semesters VII and VIII. (As per Annexure IV)

Additional Courses for MDM and Honors with Research B.Tech. (Computer Science and Engineering with Multidisciplinary Minor)

Course Category	Course Code	Course Title	Nature of Course	No of Credits	Teaching (Contact hrs/ week)		Evaluation Scheme (Marks)			Minimum Passing (Marks)			Semester
					L	P	Internal	External	Total	Internal	External	Total	
Project	CSB24RPJ401	Research Project Part I	Project	8	-	16	50	100	150	-	40	60	VII
Project	CSB24RPJ451	Research Project Part II	Project	10	-	20	50	100	150	-	40	60	VIII
		TOTAL		**18**	**-**	**36**	**100**	**200**	**300**				

Syllabus
Semester –I

Course Code: APS21BSL101 **Course Name:** Single and Multivariable Calculus **Course Category:**

Credits: 4 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:
1. To provide the basic tools of calculus for the purpose of modelling the engineering problems mathematically and obtaining solutions.
2. To convey a sense of the utility of calculus and develop technical competence of the student.

Course Outcomes:
On completion of the course, the student should be able to:
CO1. Compute derivatives in engineering problems.
CO2. Compute definite integrals arise in the problems such as arc length, surface of revolution, Work and Fluid Forces.
CO3. Find the partial derivatives and apply the knowledge of partial differentiation to find maxima and minima of functions, Jacobians, estimating error and approximation.
CO4. Calculate Area, Volume, Center of mass and Gravity using Double and Triple integral.
CO5. Solve the problems on fundamental theorems of vector calculus such as Green's, Stokes and Divergence theorems.

Contents:

Unit	Content	Teaching Hours
1	**Differentiation and its Applications:** Limit ,Continuity and Differentiation, Rate of Change in sciences and Engineering, Chain Rule and implicit differentiation, Related rates, Extreme value theorem, Rolle's Theorem, Lagrange's Mean value theorems, Nth derivatives, Taylor and Maclaurin series Expansions, Linear approximations and Differentials.	10
2	**Integration and its Applications:** Integration, Reduction formulae, Beta and gamma function, Properties, Evaluation of integrals using Beta and gamma functions, Application of Definite integrals to volume, arc length, surface of revolution, Work and Fluid Forces.	10
3	**Partial Differentiation and its Applications:** Limit and Continuity, Partial derivatives of first and higher orders, Total differentials, Errors and Approximations, Total derivative, Extreme values and saddle points, Method of Lagrange multipliers, Jacobians, Vector differentiation, Gradient, Curl and Divergence, directional derivatives.	10
4	**Multiple Integrals and its Applications:** Double integrals ,Evaluation of Double integrals, Change of order of	10

	integration in double integrals,Change of variables (Cartesian to polar), Applications: Area and Volume, Root mean square value, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian),Volume by triple integration.	
5	**Vector Integration and its Applications:** Line integrals, Surface integrals, Green's Theorem, Stokes Theorems, Divergence theorems.	10

Text Books:

1. James Stewart, Calculus Early Transcendental, 7th edition, Cengage.

2. George B. Thomas, Ross L. Finney, Calculus and Analytical Geometry, 9th edition, Pearson.

3. Howard Anton, Irl Bivens, Stephens Davis, Calculus, 10th Edition, Wiley.

Reference Books:

1. Shanti Narayan, Differential Calculus, S. Chand & Co.

2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, New York.

3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi.

4. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II) , Pune Vidyarthi Griha Prakashan, Pune.

5. H. K. Das and Rajnish Verma, Higher Engineering Mathematics , S. Chand & CO. Pvt. Ltd., New Delhi.

6. K. D Joshi, Calculus for Scientists and Engineers, CRC Press.

7. Prasad and Reena Garg, Advanced Engineering Mathematics, Khanna Publishing Company Private Limited, New Delhi.

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Course Code: APS21BSL102 **Course Name:** Engineering Physics **Course Category:**

Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Student should know Basic Physics and basic Mathematics

Course Objectives:

1. To impart knowledge in basic concepts of physics relevant to engineering applications
2. To introduce advances in technology for engineering applications.

Course Outcomes:

On completion of the course, the student should be able to:

CO.1 Summarize fundamentals of electron optics, modern physics and ultrasonic waves related to the engineering fields.

CO.2 Identify the importance of the optical phenomenon i.e. interference, diffraction and polarization in relevance with its engineering applications.

CO.3 Classify the material on the basis of electric conductivity as semiconductor and superconductors and dielectric materials this leads to their fascinating applications.

CO.4 Recognize the use of laser and optical fibers in various fields.

CO.5 Outline basics of crystallography and X- rays and demonstrate the applications of nano-materials relevant to engineering program.

Contents:

Unit	Content	Teaching Hours
1	**Modern Physics:** Electron Optics: e/m by Thomson's method, Positive ray, Bainbridge mass spectrograph. Quantum Mechanics: Role and concepts, De- Broglie's hypothesis, Uncertainty Principle, Fundamentals of quantum computing, Quantum features Ultrasonic Waves: Production of ultrasonic waves (Magnetostriction & Piezoelectric method), Applications. Numericals.	8
2	**Wave Optics:** Interference- Interference in thin films (reflected light), Newton's Rings, Engineering applications of Interference. Diffraction- Fresnel's and Fraunhofer Diffraction, Theory of plane transmission Grating. Polarization-Polarization by reflection and double refraction, Optical activity, Specific rotation, Construction and working of Laurent's half shade polarimeter, Engineering applications of Polarization. Numericals.	8
3	**Materials of Technological Importance:** Dielectric Materials: Introduction, Types of polarizations: Electronic and Ionic, Orientation Polarizations - Applications of Dielectrics Semiconducting Materials: Introduction, Fermi energy in Intrinsic	8

	semiconductors and extrinsic semiconductors, Hall effect, Applications of Semiconductors. Numericals. Superconducting Materials: Introduction, Type – I and Type – II superconductors, Meissner effect, BCS Theory, Application	
4	**Optoelectronic Materials and Devices:** LASER : Absorption, spontaneous and stimulated emission, population inversion n, pumping mechanism, Construction and working of Ruby laser, Construction and working of He-Ne laser. Lasers in various technological applications. Introduction to Optical Fibers-Introduction Acceptance Angle-Numerical Aperture, Applications of optical fibers. Numericals.	8
5	**Physics of Materials:** Crystal structure: Unit cell, Coordination number, atomic radius, packing density of cubic system. X-rays: Bragg's law, X-Ray Diffraction (XRD), Industrial Applications of X-Rays. Numericals. Particle detector: G.M. Counter Nano-Materials: Basic principles of nano science and technology, properties, applications of nanotechnology.	8

Text Books:

- Engineering Physics- H.K. Malik & A.K. Singh, McGraw Hill publication.
- Engineering Physics - R.K. Gaur and S. L. Gupta. Dhanpat Rai Publications Pvt. Ltd.-New Delhi
- M. N.Avadhanulu, P.G.Kshirsagar "A Text book of Engineering Physics"-S. Chand Publications.
- B.K. Pandey and S. Chaturvedi, Engineering Physics, Cengage Learning, 2012.

Reference Books:

1. Fundamental of Physics - Halliday and Resnik. Willey Eastern Limited.
2. Introduction to Electrodynamics –David R. Griffiths.
3. Concept of Modern Physics – Arthur Beizer. Tata McGraw-Hill Publishing Company Limited.
4. Optics – AjoyGhatak.MacGraw Hill Education (India) Pvt. Ltd.
5. Science of Engineering Materials- C.M. Srivastava and C. Srinivasan. New Age International Pvt.Ltd.
6. Solid State Physics – A.J. Dekker. McMillan India –Limited.
7. The Feynman Lectures on Physics Vol I, II,III.
8. Introduction to solid state physics – Charles Kittel. John Willey and Sons
9. T Pradeep "A Text book of Nano Science and Nano Technology"-Tata Mc GrawHill 2019.

E-Resources:

1. https://nptel.ac.in/courses/122107035/ Engineering Physics
2. https://youtu.be/98xoZknQjl8 Polarization
3. https://youtu.be/yINtzw63KncMaxwell's Equations and EM theory
4. https://youtu.be/bwreHReBH2AMaxwell's Equations and EM theory

5. https://www.youtube.com/playlist?list=PLuv3GM6-gsE3-hVNaw-YEb7EeY5XVPZdz
6. Maxwell's Equations and EM theory(nptel)
7. https://nptel.ac.in/courses/115105120/ Experimental Physics
8. https://youtu.be/2CsMpEBl5QY Crystal Structure and X- rays
9. https://youtu.be/z_8aJPLr21E Crystal Structure and X- rays
10. https://youtu.be/_Ckh-60B6LY Condensed matter Physics
11. https://youtu.be/QQZ6EGf0Ju8 Magnetic Properties
12. https://youtu.be/DDLljK1ODeg Magnetic Materials
13. https://youtu.be/etjZmdmrjSU Dielectrics
14. https://youtu.be/k6ZxP9Yr02E Semiconductor
15. https://youtu.be/D-9M3GWoBrw Superconductivity
16. https://youtu.be/GglT1RoBPzg Superconductivity
17. https://youtu.be/VHp2Ff5N_bs Superconductivity
18. https://youtu.be/FNp81kkxj5c LASER
19. https://youtu.be/YvrwVK9ZqQY LASER
20. https://nptel.ac.in/courses/115107095/ Optic Fiber
21. https://youtu.be/cjBPnIXK60U Quantum Mechanics (Prof.H.C. Verma)
22. https://youtu.be/BDuqChhUhM0 Divergence and Curl(Prof.H.C. Verma)
23. https://youtu.be/sCviGSMaYfI Divergence and Curl (Prof.H.C. Verma)
24. https://youtu.be/SZCsFS9izfQ Divergence and Curl

And other related videos from following resources

25. www.nptel.ac.in ;www.swayam.gov.in;https://inflibnet.ac.in/
26. www.sciencedirect.com
27. http://vlabs.iitb.ac.in/vlab/
28. www.youtube.com

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Course Code: APS21ESL101 **Course Name:** Python Programming **Course Category:**

Credits: 3 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

1. **Pre-requisites:** Basic Computer Knowledge & Knowledge of any programming language(optional)

Course Objectives:

1. To understand fundamental concepts in Python Programming
2. To learn the different Conditional Loops and Iteration
3. To understand various data structures and packages

Course Outcomes:

CO1: Describe programming fundamentals of python.
CO2: Interpret the python syntax and semantics of control flow statements.
CO3: Identify the methods to create and manipulate programs with python data structures.
CO4: Use modular approach for problem solving.
CO5: Apply advanced features and packages of python programming required for data science.

Contents:

Unit	Content	Teaching Hours
1	**Python for everybody:** Why Program, Hardware Overview, Python as a Language, Why Python, Installation Python Jupyter notebook, Using the Python Playground ,how to write program and compile in Jupyter notebook. Writing input and output statements in Python, output formatting, Writing comments, keywords in Python .variables and variable assignments, Operators ,writing expressions	6
2	**Conditional Loops and Iteration:** Conditional Statements, in Python, Loops and Iteration, Definite Loops, Finding the Largest Value, Loop Idioms	6
3	**Data Structures in Python:** Strings, Manipulating Strings, Files, Processing Files, Dictionaries, sets, Tuples, Lists, Manipulating Lists, Lists and Strings, Strings, Manipulating Strings	6
4	**Functions, Modules and Packages:** Functions, Lambda functions, Recursive function, Types of functions, modules and packages.	6
5	**Packages in Python for Data Science:** Numpy introduction, Numerical operations on Numpy, Introduction of Matplotlib, getting started with Pandas, Data frames basics in Pandas, key operations on data frames. Introduction to Data Science, Binary search, finding elements in common in lists using Hash, Finding largest elements, Introduction of SQL	6

Text Books:

1. Kent D. Lee, “Python Programming Fundamentals”, Second Edition, Springer Publication.

2. Wes McKinney, “Python for Data Analysis” O’Reilly Publication.

Reference Books:

1. The Python Language Reference: http://docs.python.org/2/reference/index.html

2. The Python Standard Library: http://docs.python.org/2/library/

3. https://docs.scipy.org/doc/scipy/reference/tutorial/stats.html

4. http://matplotlib.org/api/mlab_api.html#module-matplotlib.mlab

5. http://conference.scipy.org/proceedings/scipy2010/pdfs/seabold.pdf

6. http://seaborn.pydata.org

7. https://www.datacamp.com/community/data-science-cheatsheets

8. PEP 20 -- The Zen of Python: https://www.python.org/dev/peps/pep-0020/

9. https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html

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General Instructions:

The theory classes are to be conducted batch wise in Lab. Each class should be divided into four batches.

Course Code: APS21SL102 **Course Name:** Engineering Graphics **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:

1. Understand the basic principles of engineering graphics and improve the visualization skills
2. To gain knowledge on projection of points, straight lines, planes, solids.
3. To understand the real life objects through Drawings.
4. To know the principles of orthographic and isometric projections.

Course Outcomes:

CO 1. Identify basic concepts in drawing and its application.
CO 2. Plan and prepare neat orthographic drawings of points, straight lines, planes and solids.
CO 3 To visualize and draw orthographic and isometric projection of solids.
CO 4. Acquire skill to draw real life engineering objects by using the engineering drawing.

Contents:

Unit	Content	Teaching Hours
1	**Projections of Straight Lines:** Introduction to Engineering Graphics, Need of Engineering Drawing, Drawing Instruments, BIS code of practice for general engineering drawing, Projections of Points in Four Quadrants, Projections of Points in Reference Plane, Line Parallel to both the Plane, Line Parallel to One Plane and Perpendicular to the other, Line Inclined to One Plane and Parallel to The Other, Line Inclined to Both the Reference Planes, Traces of Line (Only first quadrant to be considered)	5
2	**Engineering Curves:** Curves used in Engineering Practice, Conic sections, Construction of conics by different methods, Rectangular-hyperbola, Cycloidal curves, Epi and hypo-cycloids, Involutes and Archimedean spiral.	5
3	**Projections of Planes:** Plane with Surface Parallel to One Plane and Perpendicular to other, Plane Inclined to One Plane and Perpendicular to other, Projections of Planes Inclined to both the Planes	5
4	**Projections of Solids:** Introduction to Solids: Prisms, Pyramid, Cylinder, Cone, Cube, Tetrahedron, Sphere, Projections of above Solids with Axis inclined to one plane, Projections of above solids with Axis inclined to both the Planes, Projection of composite solids (different arrangement of Spheres with above Solids).	5
5	**Orthographic Projections:** Orthographic projections of different Machine Parts (First Angle Projection method only)	5
6	**Isometric Projections ,Introduction to CAD:** Introduction to Pictorial views, Converting Orthographic Projections into Isometric Projections and Isometric views. Listing the computer technologies	5

	that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects	

Text Books:

1. N. D. Bhatt and Panchal V. M., “Engineering Drawing”, Charotar Publishing House, Anand
2. P. J. Shah , “ A Text Book of Engineering Drawing”, S.Chand, New Delhi
3. Shah M.B. & Rana B.C, “Engineering Drawing & Computer Graphics”, Pearson Publications,
4. Agrawal B.& Agrawal C.M, “Engineering Graphics”, TMH Publications
5. Narayana K.L. & P. Kannaiah, “Engineering Drawing”, Scitech Publications
6. P.I Vargese, “Engineering Graphics”, Mcgraw Hill Publications
7. D.A.Hindoliya,” Engineering Graphics”, B. S. Publications

Reference Books:

1.Dabhade M. L., “Engineering Graphics”, Vol.-I and Vol.-II, Vision Publications, Pune
2.K.Venugopal, "Engineering Drawing and Graphics" ,New Age International Publishers

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Course Code: MGM54AEL101 **Course Name:** Communicative English **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Basic knowledge of English

Course Objectives:
The course aims at grooming the professional ethics of the students through various personality traits and behavioral patterns focusing on communication skills.

Course Outcomes:

CO.1 communicate formally with enhanced communication Competency

CO.2 to adapt professional nonverbal communication

CO.3 construct English formal syntax and apply corporate vocabulary in written and verbal communication

CO.4 acquire listening and drafting skills with professional competency

Contents:

Unit	Content	Teaching Hours
1	**Communication andCommunication Process:** Introduction to Communication, Forms and functions of Communication, Barriers to Communication and overcoming them, Ways of Effective Communication.	5
2	**Non-verbal Communication And its types:** Kinesiccs, Occulesics, Appearance, Proxemics, Chronemics, Paralanguage, Qualities of effective speech	3
3	**English Grammar:** Overview of basic Mid-level grammar, Tenses& concept of time, Sentence construction, Corporate vocabulary, Difference between formal and informal sentences, phrases and words	3
4	**Listening Skills and Writing Skills:** Listening : Active and Passive Listening writing styles layouts BusinessLetters- job application, resignation, resume	4

Text Books/ Reference Books:

1 Ashraf Rizvi, Communication Skills for Engineers, Tata McGraw Hill

2 Sanjay Kumar, Pushp Lata, Communication Skills, Oxford University Press, 2016.

3 Meenakshi Raman, Sangeeta Sharma, Communication Skills, Oxford University Press, 2017.

4 Teri Kwal Gamble, Michael Gamble, Communication Works, Tata McGraw Hill Education,2010.

5 Anderson, Kenneth. Joan Maclean and Tossny Lynch. Study Speaking: A Course in Spoken English for Academic Purposes. Cambridge: CUP, 2004.

6 Bellare, Nirmala. Reading Strategies. Vols. 1 and 2. New Delhi. Oxford University Press,1998

7 Bhasker, W. W. S & Prabhu, N. S.: English through Reading, Vols. 1 and 2. Macmillan,1975

8 BoveeCourtland,L and Thrill, John V. Business Communication, Today McGraw Hill, NewYork, Taxman Publication (1989).

9 Murphy, Raymond. Essential English Grammar, Cambridge: University Press (2000)

10 Hewings Martin Advanced English Grammar Cambridge: University press(2003)

11 Bansal Harrison. Spoken English

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Course Code: APS21VSP101 **Course Name:** Engineering Exploration **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:

1. Understand the principles of Engineering Design and apply them to identify product requirements, objectives, and constraints through interactions with clients, users, and relevant stakeholders.
2. Develop critical thinking and problem-solving skills to analyze existing technologies, conduct surveys, study literature, and generate creative design concepts using various means and combinations.
3. Acquire proficiency in using appropriate tools and techniques to generate electronic and mechanical simulations, schematic diagrams, and product concept sketches or CAD models for effective communication of design ideas.
4. Demonstrate competency in categorizing inputs and outputs of systems in terms of materials, information, and energy, enabling a holistic understanding of engineering projects.
5. Cultivate effective teamwork and communication skills to collaborate with peers, mentors, and team members during the development and presentation of engineering prototypes.
6. Develop the ability to evaluate, defend, and communicate design decisions, progress, and project outcomes, showcasing awareness of course concepts and their application in real-world scenarios.

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Course Outcomes:

On completion of the course, the student should be able to:

LO 1. Effectively interact with clients, users, and stakeholders to gather relevant information and derive product requirements, thereby demonstrating proficiency in Requirement Analysis and Client Interaction.

LO 2. Analyse and compare existing technologies, conduct surveys, and study literature to identify potential solutions, and develop objective trees and function trees, reflecting their competency in Conceptual Design and Analysis.

LO 3. Create product concept diagrams, combining different means, and prepare sketches or CAD models to present their design ideas aesthetically and coherently, indicating proficiency in Product Concept Development.

LO 4. Categorize system inputs and outputs in terms of materials, information, and energy, illustrating their understanding of System Categorization and Analysis.

LO 5. Exhibit hands-on skills in physical assembly, connection, and demonstration of engineering prototypes, showcasing Proficiency in Prototyping and Implementation.

LO 6. Explain and defend their design choices, project progress, and outcomes during presentations, demonstrating Communication Skills, Project Evaluation, and Awareness of Engineering Concepts.

Contents:

Sr. No.	List of Practical	Lab Hours
1	The course is conducted in the following modules: 1. Introduction to Engineering Exploration 2. Engineering Design 3. Platform Based Development 4. Mechanisms 5. Data acquisition and analysis 6. Engineering Ethics 7. Project Management	10
2	The following practical contents are delivered in an integrated mode along with theory: 1. Conceptualizing a product 2. Designing a product with constraints 3. Simulation of electronic circuits (at least 10 sets) 4. Implementation of electronic circuits (at least 5 sets) 5. Implementation of four bar chain mechanism 6. Conversion of problem statement to need statement 7. Identification of objectives, constraints and functions 8. Generation of black box, glass box and expanded class box 9. Generation of morphological chart 10. Generation of concepts 11. Comparison of concepts 12. Selection of concept 13. Implementation and testing of prototype	20

Reference Books:

- George E. Dieter and Linda C. Schmidt (2009), Engineering Design, 4ed, Mc Graw Hill Higher Education
- Clive L. Dym, Patrick Little, and Elizabeth J. Orwin (2014), Engineering Design: A project-based introduction, 4ed, John Wiley and Sons
- G. Pahl, W. Beitz, J. Feldhusen and K.-H. Grote (2007), Engineering Design: A systematic approach, 3ed, Springer, New York

Course Code: APS21BSP101 **Course Name:** Engineering Physics Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Student should know the basic aspects of measurements like least count and range of instrument, scale identification, accuracy, error etc.

Course Objectives:
The Objective of this course is
1. To make the students gain practical knowledge to co-relate with the theoretical studies.
2. To achieve perfectness in experimental skills.
3. The study of practical applications will bring more confidence.

Course Outcomes:
At the end of the laboratory course, the students will be able to,
LO1: Plot the I-V characteristics of pn junction diode and determine the value of band gap energy which can be used for core engineering Courses.
LO2: Perform optical experiments; using light properties like interference, polarization, diffraction and their applications.
LO3: Interpret the results and analyze the data and use the experimental data to plot the graph for a best fit.
LO4: Discuss the characteristics of plateau region and determine operating voltage of G.M. counter
LO5: Determine the numerical aperture and bending loss of optical fibre cable

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Contents:

Sr. No.	List of Practical	Lab Hours
1	Thomson's Bar Magnetic method -Determination of e/m of electron	02
2	Newton's rings -Determination of radius of curvature of Plano convex lens.	02
3	Polarization - Half shade Polarimeter -Determination of specific rotation of optically active material.	02
4	Diffraction - Determination of wavelength of light by plane transmission grating.	02
5	Wedge Shaped film -Determination of thickness of thin wire	02
6	Semiconductors – To study of forward and reverse bias characteristic of Semiconductors diode	02
7	Fibre Optics – Determination of Bending losses in optical fibre	02
8	Fibre Optics – Determination of Numerical Aperture of given optical fibre	02
9	Determination of operating voltage of G.M. tube and characteristics of plateau region	02
10	Determination of wavelength of laser source	02

Course Code: APS21ESP101 **Course Name:** Python Programming Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The Objective of this course is

Course Outcomes:
LO1 : Demonstrate python program using development environment.
LO2: Develop logical thinking to solve the problems using programming fundamental concepts.
LO3 : Construct python program using various data structures.
LO4 : Apply modularization approach for solving complex problem.
LO5: Make use of various packages in Python for data science.
LO6 : Implement different SQL commands in python.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Program to perform input/output operations Write a program to take input (integer, float, string) and print it.	02
2	Program based on operators 1.Write a program to simulate a simple calculator (+ - / * %) that takes two operands as input and displays the result 2.Write a program to find area and perimeter of geometric objects. 3.The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches and centimeters. 4. Write a Program to interchange two numbers. 5.Write a program to compute Fahrenheit from centigrade	02
3	Programs based on Decision making. 1.Write a program to read marks from keyboard and your program should display equivalent grade according to following table(else-if) (ladder) Marks Grade 100 - 80 Distinction 79 - 60 First Class 59 - 40 Second Class < 40 Fail 2.Write a program to input basic salary of an employee and calculate gross salary according to given conditions. Basic Salary <= 10000 : HRA = 20%, DA = 80% Basic Salary is between 10001 to 20000 : HRA = 25%, DA = 90% Basic Salary >= 20001 : HRA = 30%, DA = 95% 3.If the ages of three brothers are input through the keyboard, write a C Program to determine the youngest and oldest of the three. 4.Write a program to calculate overtime pay of employee. Overtime is paid at	02

	the rate of Rs. 12.00 per hour for every hour worked above 40 hours. Assume that employee do not work for fractional part of an hour. 5.Write a program for checking the speed of drivers. If speed is less than 70, it should print "Ok". Otherwise, for every 5km above the speed limit (70), it should give the driver one demerit point and print the total number of demerit points. For example, if the speed is 80, it should print: "Points: 2". If the driver gets more than 12 points, the function should print: "License suspended"	
4	Programs using while and for loops 1.WAP to find factorial of given number 2.WAP to check whether given number is Palindrome or not 3.WAP to check whether given number is Armstrong or not 4.WAP to print Fibonacci series 5.Write a Python program which iterates the integers from 1 to 50. For multiples of three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz". 6.WAP to check whether given number is Perfect number or not 7.WAP to check whether given number is Prime number or not 8.Write C program to print given star and number patterns and reverse it. • * 1 • ** 12 • *** 123 **** 1234	02
5	Programs on string 1.Write Python Program to find length of string without using len() function. 2.Count all letters, digits, and special symbols from a given string. 3.Python Program to Count the Number of Vowels in a String. 4.Python Program to Calculate the Number of Upper Case Letters and Lower Case Letters in a String. 5.Python Program to Check whether given string is palindrome or not	02
6	Programs on List and Tuple 1.Write a Python program to sum all the items in a list. 2.Write a Python program to multiply all the items in a list 3.Write a Python program to get the largest number from a list. 4.Write a Python program to get the smallest number from a list 5.Write a Python program to count all elements in list and count Occurrences Of A List Item In Python 6.Write a Python program to create a tuple with different data types 7. Write a Python program to check whether an element exists within a tuple 8.Write a Python program to reverse a tuple 9. Write a Python program calculate the product of all the numbers given in tuple. Original Tuple: (2, 4, 8, 8, 3, 2, 9) Product - multiplying all the numbers of the said tuple: 27648	02
7	Programs on set and dictionary 1.Write a Python program to concatenate following dictionaries to create a new one. Sample Dictionary : dic1={1:10, 2:20} dic2={3:30, 4:40} dic3={5:50,6:60}	02

	Expected Result : {1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60} 2.Write a Python program to check whether a given key already exists in a dictionary 3.Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (x, x*x) Sample Dictionary (n = 5) : Expected Output : {1: 1, 2: 4, 3: 9, 4: 16, 5: 25} 4.Write a Python program to merge two Python dictionaries 5.Write a Python program to get the maximum and minimum value in a dictionary 6.Write a Python program to create set difference, Union and intersection 7.Write a Python program to check if two given sets have no elements in common	
8	Programs using function 1. Write Functions to calculate your trip's costs: 2. Define a function called hotel_cost with one argument nights as input 3. Define a function called plane_ride_cost that takes a string, city, as input. 4. Define a function called rental_car_cost with an argument called days. 5. Define a function called trip_cost that takes two arguments, city and days. Like the example above, have your function return the sum of calling the rental_car_cost(days), hotel_cost(days), and plane_ride_cost(city) functions. 6. Write a program in to check a given number is even or odd using the function. 7. Write a function Exchange to interchange the values of two variables, say x and y. illustrate the use of this function in a calling function. 8. Write a program to find Sum of natural number using recursion. 9. Write a program to print Fibonacci series number using recursion	02
9	Program using NumPy, Matplotlib and Pandas library 1.Write a program to perform matrix addition, substraction, multiplication. 2.Plot all types of graph using Matplotlib. 3.Write a program which perform basic operation of Pandas.	02
10	Program on SQL Commands 1. Write a program of binary search 2. Write a program which perform basic SQL commands 3. Programs based on real life problems/GUI based programs	02

Course Code: APS21ESP102 **Course Name:** Engineering Graphics Lab **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The Objective of this course is

Course Outcomes:
Students will be able to,
1. Develop competence in correct expression of the visualized objects
2. Dimension and annotate two-dimensional engineering drawings
3. Plan and prepare neat orthographic drawings of points, straight lines, planes and solids
4. Develop the ability to visualize and draw orthographic and isometric projection of solids

Contents:

Sr. No.	List of Practical	Lab Hours
1	Drawing three problems based on projections of lines on half imperial size drawing sheet	02
2	Drawing three problems based on engineering curves on half imperial size drawing sheet	02
3	Drawing three problems based on projections of planes on half imperial size drawing sheet	02
4	Drawing three problems based on projections of solids on half imperial size drawing sheet	02
5	Drawing three problems based on orthographic projections on half imperial size drawing sheet	02
6	Drawing three problems based on isometric projections on half imperial size drawing sheet	02
7	Demonstration of CAD software in CAD lab, drawing simple objects using various commands	02

Course Code: APS21ESP103 **Course Name:** Recent Trends in Integrated Technologies Lab
Course Category:

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

1.To introduce students the basics of additive manufacturing/rapid prototyping and its applications in various fields, reverse engineering techniques.
2.To recognize industrial control problems suitable for Industrial Robotics.
3.To acquire basic skills in exploring the potential of the drone technology in professional activities
4.Ability to recognize industrial automation problems suitable for PLC control.

Course Outcomes:

LO1. Prepare 3D Model (slice & print) in either Stratasys or Zortrax & generate scan data through Hexagon portable scanning arm.
LO2.Describe basic industrial robotics & it's applications
LO3.Operate a small drone in a controlled environment
LO4.Explain principles of sensor, PLC & applications.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Scan to CAD, CAD to STL conversion& patching, machine setup & processes for printing	02
2	Programming the TATA Robot(pendant) for pick & place, programming the Yaskawa Robot(pendant) for motion planning, Demonstrate welding exercise by the instructor	02
3	Introduction to components of drone, Demonstration of assembly of drone, Demonstration of mission planning & flying the drone	02
4	Controller & sensor & their interfacing, basic ladder logic instructions, pneumatic& hydraulic actuator	02

Reference Books:

1. A Step-by-Step Guide For Beginners: Aircraft Design & Construction Design Guide by :Merlin Debrie
2. Industrial Automation & Robotics By A.K. Gupta & S.K.Arora
3. Additive Manufacturing Principles, Technologies & Applications By C.P.Paul (TMH)
4. Basics of unmanned aerial vehicle By Garvit Pandya (Motion press)

Course Code: MGM54AEP101 **Course Name:** Communicative English Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Basic knowledge of English

Course Objectives:
The course aims at grooming the professional ethics of the students through variouspersonality traits and behavioural patterns focusing on communication skills.

Course Outcomes:
The Students will be able to
LO.1 Introduce themselves formally and informally through practice.
LO.2 Pronounce English vowel and Consonant sounds effectively
LO.3 Participate effectively in G.Ds, Presentations, & Interviews
LO.4 Face Interviews competently
LO.5 Draft resume, business letters, reports formally
LO.6 Comprehend the meaning of English text by comprehension techniques.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Self Introduction	02
2	Pronunciation of Vowel sounds in English	02
3	Pronunciation of consonant sounds in English	02
4	Group discussion	04
5	Presentation techniques	04
6	Interview techniques	04
7	Letter writing	04
8	Email writing, Agenda of the meeting and notices	02
9	Resume Writing	02
10	Report writing	02
11	Skimming & Scanning	02

Course Code: MGM82CCP101 **Course Name:** National Cadet Corps **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
Nil

Course Outcomes:
Nil

Contents:

Unit No.	Content	Hours
1	**NCC General, National Integration and Awareness, Social Service and Community Development, Drill:** Aims, Objectives and Organization of NCC, Duties of NCC Cadet, National Integration Importance and Necessity, Factors Affecting National Integration, Foot Drill, Drill With Arms, Ceremonial Drill with Arms, Social Service and Community Development Activities- Pollution, Mission Indradhanush, Beti Bachao Beti Padhao, Tree Plantation, Digital Awareness.	12
2	**Personality Development, Disaster Management, Weapon Training, Map Reading :** Critical and Creative Thinking, Decision Making, Self Awareness, Public Speaking, Interview Skills, Types, Org, Capability and Role of NCC cadets, Initiative Trg, Organizing Skills, Do's & Don't , Introduction and Characteristics of .22 rifle, Handling of .22 rifle, Introduction to Map Reading, Conduct of MR- Google and Tourist Maps and Apps.	10
3	**Health and Hygiene, Environmental Awareness and Conservation, Adventure, Obstacle Training:** Hygiene & Sanitation (Personal & Camp Hygiene) Soch Vichar, First Aid in Common Medical Emergencies, Treatment & Care of Wounds, Introduction Yoga & Exercises, Water Conservation, Energy Conservation, Introduction Adventure Activities. Obstacle Course.	10
4	**Leadership, Introduction to Infantry Weapons and Equipments:** Traits, Indicators, Motivation, Ethics, Case Studies- Chhatrapati Shivaji Maharaj, Maharana Pratap, Jhasi ki Rani, Ratan Tata, Narayan Murty, Rabindra Nath Tagor, Organization of Infantry Battalion and its weapons.	08
5	**Armed Forces, Field Craft and Battle Craft, :** Armed Forces, Army, CAPF, Police, Modes of Entry to Army, CAPF, Police, Introduction to Field Craft, Indication of Landmark, Observation, Camouflage and Concealment, Fire and Move Capsule.	08

References:

- Cadet's Handbook- Common Subject, all wings by DG NCC, New Delhi.
- Cadet's Handbook- Common Subject by NCC Directorate- Bhubaneshwar.
- Cadet's Handbook- Specialised Subjects, Army, Navy, Air-Force by DG NCC, New Delhi.
- NCC OTA Precise by DG NCC, New Delhi.
- Chanakya's 7 Secrets of Leadership by Radhakrishanan Pillai and D. Shivnandhan.
- National Cadets Corps (India) by Lambert M. Suvarkar.

E-Resources:

1. National Cadet Corps, Youth in Action (Google eBook).

https://indiancc.nic.in/dg-ncc-lt-gen-gurbirpal-singh/

www.youtube.com

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Course Code: MGM82CCP103 **Course Name:** Sports **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

1. To understand the importance of sports in physical and mental development.
2. To learn about the different types of sports.
3. To learn about the different types of indoor games.
4. To participate in indoor games.
5. To learn about the different types of outdoor games.
6. To participate in outdoor games.

Course Outcomes:

- Upon completion of this course, students will be able to
- Demonstrate knowledge of the history, benefits, types, equipment, and safety of sports
- Demonstrate proficiency in the basic skills of indoor and outdoor games
- Understand the rules and regulations of selected sports
- Participate in sports competitions

Contents:

Sr. No.	Content	Hours
1	Football: History of Football: The earliest forms of football can be traced back to ancient China, Greece, and Rome. In England, the game of football developed in the 19th century, with different rules being used by different schools and organizations. In 1863, the Football Association (FA) was founded, and it standardized the rules of the game. The first international match was played between England and Scotland in 1872. Football became an Olympic sport in 1900, and the first World Cup was	12

	held in 1930. Today, football is the most popular sport in the world, with billions of fans around the globe. **Fundamental Skills of Football** Dribbling: Dribbling is the ability to move the ball with your feet while keeping control of it. It is a essential skill for all footballers, as it allows you to move past defenders and create scoring opportunities. Passing: Passing is the ability to accurately and effectively throw the ball to your teammates. It is another essential skill, as it allows you to move the ball up the field and create scoring opportunities. Shooting: Shooting is the ability to kick the ball with power and accuracy. It is the most important skill for scoring goals, and it is essential for all footballers to develop a good shot. Heading: Heading is the ability to use your head to control and direct the ball. It is a valuable skill for both attacking and defending, and it is important for all footballers to learn how to head the ball effectively. Tackling: Tackling is the ability to take the ball away from an opponent. It is an important skill for defenders, but it is also valuable for midfielders and attackers.	
2	**Basket Ball** **History of Basketball** Basketball was invented by James Naismith in 1891 at the International YMCA Training School in Springfield, Massachusetts. Naismith was a physical education instructor who was looking for a game that would be less injury-prone than football. He nailed two peach baskets to the lower rail of a balcony and used a soccer ball to play the game. The first game of basketball was played on December 21, 1891, with nine players on each team. The rules of basketball have evolved over time, but the basic premise of the game has remained the same. Today, basketball is one of the most popular sports in the world, with millions of players and fans around the globe. **Fundamental Skills of Basketball** Dribble: Dribbling is the ability to move the ball with your hands while keeping control of it. It is an essential skill for all basketball players, as it allows you to move past defenders and create scoring opportunities. Passing: Passing is the ability to accurately and effectively throw the ball to your teammates. It is another essential skill, as it allows you to move the ball up the court and create scoring opportunities. Shooting: Shooting is the ability to throw the ball through the hoop with power and accuracy. It is the most important skill for scoring points, and it is essential for all basketball players to develop a good shot. Rebounding: Rebounding is the ability to catch the ball after it has been missed by a shooter. It is an important skill for both offense and defense, as it allows teams to get second chances at scoring. Defense: Defense is the ability to prevent the other team from scoring points. It is an essential skill for all basketball players, as it is impossible to win a game without playing good defense.	10

3	**Volley Ball** **History of Volleyball** Volleyball was invented in 1895 by William G. Morgan, a physical education instructor at the Young Men's Christian Association (YMCA) in Holyoke, Massachusetts. Morgan was looking for a game that would be less vigorous than basketball, and he created volleyball as a way to keep his students active during the winter months. The original name of the game was "mintonette," but it was renamed "volleyball" in 1896. Volleyball quickly spread throughout the United States and around the world, and it became an official Olympic sport in 1964. **Fundamental Skills of Volleyball** Passing: Passing is the ability to receive the ball from the opponent and direct it to a teammate. It is an essential skill for all volleyball players, as it allows the team to keep possession of the ball and start an attack. Setting: Setting is the ability to control the height and direction of the ball so that a teammate can spike it. It is a critical skill for setters, as they are responsible for setting up the team's offense. Spiking: Spiking is the ability to hit the ball over the net with power and accuracy. It is the most important skill for scoring points in volleyball, and it is essential for all attackers to develop a good spike. Blocking: Blocking is the ability to prevent the opponent from spiking the ball over the net. It is an important skill for blockers, as they can prevent the other team from scoring points. Digging: Digging is the ability to prevent the opponent from scoring a point by returning the ball over the net. It is an important skill for all defenders, as they are responsible for preventing the other team from scoring points	10
4	**Kabaddi** **History of Kabaddi** Kabaddi is a contact team sport that originated in India. It is believed to have originated in the Indian subcontinent over 4,000 years ago. The game is mentioned in the Sangam literature of Tamil Nadu, which dates back to the 3rd century BC. Kabaddi was first played as a competitive sport in the Indian Olympic Games in 1938. It was included as a demonstration sport at the 1982 Asian Games in Delhi, and it became a full medal sport in the 1990 Asian Games in Beijing. Kabaddi is now played in over 100 countries around the world. **Fundamental Skills of Kabaddi** Dabki: Dabki is the act of entering the opponent's half of the court while chanting "kabaddi, kabaddi." It is a fundamental skill for all raiders, as it allows them to enter the opponent's half of the court without being tackled. Touch: Touching an opponent is the most important skill in kabaddi. It is how raiders score points for their team. There are many different ways to	08

	touch an opponent, such as touching their arm, leg, or torso. Tackling: Tackling is the act of preventing a raider from touching an opponent. It is a fundamental skill for all defenders, as it allows them to prevent the other team from scoring points. There are many different ways to tackle a raider, such as grabbing them, pushing them, or tripping them. Stamina: Stamina is essential for all kabaddi players, as the game is very physically demanding. Players need to be able to run, jump, and tackle for long periods of time. Agility: Agility is also important for kabaddi players, as they need to be able to change direction quickly and avoid being tackled.	
5	**Badminton** **History of Badminton** The game of badminton originated in ancient Greece, China, and India. It was brought to England in the 1870s by British army officers stationed in India. The first badminton club was founded in 1873 at Badminton House, the country estate of the Duke of Beaufort. The first official all-England badminton championships for men were held in 1899, and the first badminton tournament for women was arranged the next year. Badminton became an Olympic sport in 1992. Today, badminton is a popular sport played by millions of people around the world. **Fundamental Skills of Badminton** Grip: The grip is the most important fundamental skill in badminton. It allows you to control the racket and hit the shuttlecock with power and accuracy. There are many different grips, but the most common are the forehand grip and the backhand grip. Footwork: Footwork is essential for moving around the court and positioning yourself to hit the shuttlecock. There are many different footwork drills that you can practice to improve your footwork. Racket control: Racket control is the ability to hit the shuttlecock with power and accuracy. It is important to practice hitting the shuttlecock in different directions and with different levels of power. Timing: Timing is the ability to hit the shuttlecock at the right time. It is important to practice hitting the shuttlecock at the peak of its flight. Stamina: Stamina is essential for badminton, as it is a physically demanding sport. You need to be able to run, jump, and hit the shuttlecock for long periods of time.	08
6	**Soft Tennis** **History of Soft Tennis** Soft tennis is a racquet sport that originated in Japan in the early 20th century. It was created as a less dangerous alternative to lawn tennis, as the ball used in soft tennis is made of foam rubber and does not travel as fast as a regular tennis ball. The first soft tennis tournament was held in Japan in 1921, and the sport quickly spread to other countries in Asia.	

	Soft tennis was first introduced to the United States in the 1950s, and it has since become a popular recreational sport in the country. **Fundamental Skills of Soft Tennis** Grip: The grip is the most important fundamental skill in soft tennis. It allows you to control the racquet and hit the ball with power and accuracy. There are many different grips, but the most common are the forehand grip and the backhand grip. Footwork: Footwork is essential for moving around the court and positioning yourself to hit the ball. There are many different footwork drills that you can practice to improve your footwork. Racket control: Racket control is the ability to hit the ball with power and accuracy. It is important to practice hitting the ball in different directions and with different levels of power. Timing: Timing is the ability to hit the ball at the right time. It is important to practice hitting the ball at the peak of its flight. Stamina: Stamina is essential for soft tennis, as it is a physically demanding sport. You need to be able to run, jump, and hit the ball for long periods of time. **Here are some additional fundamental skills of soft tennis:** Ball control: The ability to control the direction and speed of the ball. Serve: The ability to serve the ball accurately and with power. Volley: The ability to hit the ball before it bounces. Overhead smash: The ability to hit the ball forcefully and accurately overhand. Drop shot: The ability to hit the ball softly and precisely so that it bounces low and close to the net.	
7	**Tennis** **History of Tennis** The origins of tennis can be traced back to a 12th–13th-century French handball game called jeu de paume (“game of the palm”), from which was derived a complex indoor racket-and-ball game: real tennis. The modern game of lawn tennis was invented in England in the 1870s by Major Walter Wingfield. Wing field created a set of rules and equipment for the game, and he called it "Sphairistike". The game quickly became popular, and it was renamed "lawn tennis" in 1874. The first lawn tennis tournament was held in 1877 at the All England Club in Wimbledon, England. Tennis became an Olympic sport in 1896. Today, tennis is a popular sport played by millions of people around the world. **Fundamental Skills of Tennis** Grip: The grip is the most important fundamental skill in tennis. It allows you to control the racket and hit the ball with power and accuracy. There are many different grips, but the most common are the forehand grip and the backhand grip. Footwork: Footwork is essential for moving around the court and positioning yourself to hit the ball. There are many different footwork	

	drills that you can practice to improve your footwork. Racket control: Racket control is the ability to hit the ball with power and accuracy. It is important to practice hitting the ball in different directions and with different levels of power. Timing: Timing is the ability to hit the ball at the right time. It is important to practice hitting the ball at the peak of its flight. Stamina: Stamina is essential for tennis, as it is a physically demanding sport. You need to be able to run, jump, and hit the ball for long periods of time.	
8	**Fencing** **History of Fencing** The earliest evidence of fencing dates back to ancient Egypt, Greece, and Rome. Fencing was used as a form of training for warfare and as a way to settle disputes. The modern sport of fencing developed in Italy in the 15th century. The first fencing competition was held in 1550 in Paris. Fencing became an Olympic sport in 1896, and it has been a part of every Games since then. Fencing is now a popular sport all over the world, and there are three main disciplines: foil, épée, and sabre. **Fundamental Skills of Fencing** Footwork: Footwork is essential in fencing, as it allows you to move quickly and efficiently around the piste. There are many different footwork drills that you can practice to improve your footwork. Bladework: Bladework is the ability to use the sword effectively. There are many different bladework techniques, and you need to practice them in order to become proficient. Parrying: Parrying is the ability to deflect an opponent's attack. There are many different parrying techniques, and you need to practice them in order to become proficient. Riposte: The riposte is the counterattack that follows a parry. It is an important skill in fencing, as it allows you to score points. Mental Focus: Mental focus is also an important skill in fencing, as it allows you to stay focused on the opponent and to avoid making mistakes.	
9	**Athletics** **History of Athletics** The history of athletics can be traced back to the ancient Olympic Games, which were held in Greece from 776 BC to 393 AD. The original events included running, jumping, throwing, and wrestling. The modern Olympic Games were revived in 1896, and athletics has been a part of every Games since then. Athletics is now a global sport, with competitions held at all levels, from local to international. **Fundamental Skills of Athletics** Running: Running is the most basic skill in athletics. It is the ability to move forward quickly and efficiently. There are many different types of running, including sprinting, distance running, and middle-distance	

	running. Jumping: Jumping is the ability to move upwards from the ground. There are many different types of jumping, including high jump, long jump, and triple jump. Throwing: Throwing is the ability to propel an object through the air. There are many different types of throwing, including shot put, discus throw, javelin throw, and hammer throw. Sprinting: Sprinting is a type of running that involves short bursts of speed. Sprinters need to be able to accelerate quickly and maintain their speed for a short period of time. Distance Running: Distance running is a type of running that involves running for long distances. Distance runners need to be able to pace themselves and maintain their energy levels for long periods of time. Middle-Distance Running: Middle-distance running is a type of running that involves running for distances between 800 meters and 1500 meters. Middle-distance runners need to be able to combine speed and endurance. High Jump: High jumping is a type of jumping that involves clearing a bar that is raised progressively higher. High jumpers need to be able to generate a lot of power in their legs and have good timing. Long Jump: Long jumping is a type of jumping that involves jumping as far as possible. Long jumpers need to have good speed and coordination. Triple Jump: Triple jumping is a type of jumping that involves jumping three times in a row. Triple jumpers need to have good speed, coordination, and power. Shot Put: Shot put is a type of throwing that involves throwing a heavy ball as far as possible. Shot putters need to have good upper body strength and technique. Discus Throw: Discus throw is a type of throwing that involves throwing a disc as far as possible. Discus throwers need to have good upper body strength and technique. Javelin Throw: Javelin throw is a type of throwing that involves throwing a spear as far as possible. Javelin throwers need to have good upper body strength and technique. Hammer Throw: Hammer throw is a type of throwing that involves throwing a heavy ball on a chain as far as possible. Hammer throwers need to have good upper body strength and technique.	
10	**Kho-Kho** **History of Kho-Kho** Kho-Kho is a tag game that originated in India. It is believed to have originated in the Indian subcontinent over 4,000 years ago. The game is mentioned in the Sangam literature of Tamil Nadu, which dates back to the 3rd century BC. Kho-Kho was first played as a competitive sport in the Indian Olympic Games in 1938. It was included as a demonstration sport at the 1982 Asian Games in Delhi, and it became a full medal sport in the 1990 Asian Games in Beijing. Kho-Kho is now played in over 100 countries around the world.	

	Fundamental Skills of Kho-Kho Touch: Touching an opponent is the most important skill in Kho-Kho. It is how raiders score points for their team. There are many different ways to touch an opponent, such as touching their arm, leg, or torso. Dive: Diving is a fundamental skill for all Kho-Kho players. It allows players to avoid being touched by the opponents. There are many different types of dives, such as front dive, side dive, and back dive. Stamina: Stamina is essential for all Kho-Kho players, as the game is very physically demanding. Players need to be able to run, jump, and dive for long periods of time. Agility: Agility is also important for Kho-Kho players, as they need to be able to change direction quickly and avoid being touched by the opponents. Teamwork: Teamwork is essential for Kho-Kho, as it is a team sport. Players need to be able to work together to score points and defend their territory.	
11	**Cricket** **History of Cricket** The history of cricket can be traced back to the 16th century in England. The game is believed to have originated from a game called "stoolball", which was played by children in the 15th century. The first recorded cricket match was played in 1611 between two teams of Kentish cricketers. Cricket became a popular sport in England during the 18th century, and it was first played in Australia in 1826. Cricket became an international sport in the 19th century, and the first Test match was played between England and Australia in 1877. Cricket is now played in over 100 countries around the world. **Fundamental Skills of Cricket** Batting: Batting is the act of hitting the ball with a bat. It is the most important skill in cricket, as it is how runs are scored. There are many different batting techniques, such as the defensive technique and the attacking technique. Bowling: Bowling is the act of delivering the ball to the batsman. There are many different bowling techniques, such as the fast bowling technique and the spin bowling technique. Fielding: Fielding is the act of catching the ball and preventing the batsman from scoring runs. It is an important skill for all cricketers, as it helps to prevent the other team from scoring runs. Running: Running is essential for scoring runs in cricket. Players need to be able to run quickly between the wickets to score runs. Stamina: Stamina is essential for all cricketers, as the game is very physically demanding. Players need to be able to run, jump, and field for long periods of time. Agility: Agility is also important for cricketers, as they need to be able to change direction quickly and avoid being run out. Teamwork: Teamwork is essential for cricket, as it is a team sport. Players need to be able to work together to score runs and prevent the other team from scoring runs.	

12	**Rifle Shooting** **History of Rifle Shooting** The history of rifle shooting can be traced back to the 16th century in Europe. The first recorded rifle shooting competition was held in 1533 in Zurich, Switzerland. Rifle shooting became a popular sport in Europe during the 18th century, and it was first introduced to the United States in the 1770s. Rifle shooting became an Olympic sport in 1896, and it has been a part of every Games since then. Rifle shooting is now a popular sport all over the world. **Fundamental Skills of Rifle Shooting** Accuracy: Accuracy is the most important skill in rifle shooting. It is the ability to hit the target with the bullet. There are many different factors that affect accuracy, such as the stance, the grip, the breathing, and the trigger control. Consistency: Consistency is also important in rifle shooting. It is the ability to hit the target with the same accuracy shot after shot. There are many different factors that affect consistency, such as the mental focus and the physical preparation. Stance: The stance is the position of the body when shooting. It is important to have a stable stance in order to be accurate. There are many different stances that can be used, such as the standing stance, the kneeling stance, and the prone stance. Grip: The grip is the way that the rifle is held. It is important to have a firm grip in order to be accurate. There are many different grips that can be used, such as the weaver grip and the isosceles grip. Breathing: Breathing is important in rifle shooting because it can affect the accuracy of the shot. It is important to breathe slowly and evenly before and after the shot. Trigger Control: Trigger control is the ability to pull the trigger smoothly and evenly. It is important to avoid jerking the trigger, as this can cause the shot to go off target.	
13	**Yoga** **History of Yoga** Yoga is a mind and body practice with a 5,000-year history in ancient Indian philosophy. The word "yoga" comes from the Sanskrit word "yuj," which means "to yoke" or "to unite." Yoga is a system of physical postures, breathing exercises, and meditation designed to help practitioners achieve physical, mental, and spiritual well-being. The earliest written records of yoga date back to the 2nd century BCE, and the practice has been evolving ever since. Yoga has spread to all corners of the world, and there are now many different styles of yoga practiced today. **Fundamental Skills of Yoga** Postures: The postures, or asanas, are the physical component of yoga. There are many different postures, and they can be practiced in a variety	

	of ways. Breathing: Breathing, or pranayama, is an important part of yoga. There are many different breathing techniques, and they can be used to help relax the body and mind. Meditation: Meditation is the mental component of yoga. There are many different meditation techniques, and they can be used to help focus the mind and achieve a state of peace.	
14	**Swimming** **History of Swimming** Swimming is one of the oldest sports in the world, with evidence of swimming dating back to 2500 BC. The earliest recorded swimming competitions were held in ancient Greece and Rome. Swimming became an Olympic sport in 1896, and it has been a part of every Game since then. Swimming is now a popular sport all over the world, and there are many different types of swimming, including freestyle, backstroke, breaststroke, butterfly, and individual medley. **Fundamental Skills of Swimming** Breathing: Breathing is one of the most important skills in swimming. It is important to be able to breathe efficiently while swimming in order to avoid getting tired. Body Position: Body position is another important skill in swimming. It is important to maintain a good body position in order to be hydrodynamic and to swim efficiently. Stroke Technique: Stroke technique is the way that the arms and legs are used to propel the body through the water. There are many different stroke techniques, and it is important to develop a good stroke technique in order to swim efficiently. Drills: Drills are exercises that can be used to improve swimming skills. There are many different drills, and they can be used to improve different aspects of swimming, such as breathing, body position, and stroke technique. Mental Toughness: Mental toughness is also an important skill in swimming. It is important to be able to stay focused and motivated during long swims, especially in competitions.	

Semester –II

Course Code: APS21BSL103 **Course Name:** Linear Algebra and Differential Equations **Course Category:**

Credits: 4 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:

1. This course aims to make the students become familiar with the basic concepts of linear algebra with a thorough understanding of vector spaces, linear transformations and matrix operations enhancing the students' ability to reason mathematically and able to apply this knowledge to many fields in engineering, statistics and computer science.

2. Create and analyze mathematical models using differential equations.

Course Outcomes:

On completion of the course, the student should be able to:

1. Understand basic concepts such as vector spaces, linear dependence / independence of vectors, basis and linear maps.

2. Analyze and calculate eigen values, eigen vectors, rank nullity of a matrix / linear map.

3. Prove theorems, apply Gram-Schmidt process on inner product spaces, diagonalize special matrices.

4. Solve ordinary differential equations of first order and apply knowledge of differential of equations to solve engineering problems.

5. Find the solution of linear differential equations having their applications in mechanical and electrical systems.

Contents:

Unit	Content	Teaching Hours
1	**Matrices and Vector Spaces:** Basic properties of matrices, row operations and Gauss elimination, Inverse of a matrix. Basic concepts in linear algebra: vector spaces, subspaces, linear independence and dependence of vectors, bases, dimensions. Row and Column spaces, rank. Applications to systems of linear equations, Inverse transformation.	10
2	**Linear mappings and Diagonalization:** Linear mappings, representation by matrices, rank-nullity theorem, Diagonaliazation, Eigen values, Eigen vectors and their basic properties, Cayley Hamilton Theorem.	10
3	**Inner Product Spaces and Quadratic Forms:** Inner Product Spaces , Orthogonality, Gram-Schmidt process, Geometric Applications of Linear Transformation, Quadratic Forms: Positive Definiteness and applications	10

4	**First order ordinary differential equations and Applications:** Exact , Linear ,Bernoulli and separable differential equations, Applications to Population growth/decay, Mixing problems, Draining tank/Torricelli's Law problems, Newton's Law of Cooling, Electric circuits, Falling bodies.	10
5	**Ordinary differential equations of higher orders and Applications:** Linear differential equations with constant and variable coefficients, method of variation of parameters, Applications to mass spring systems and electrical circuits and Bending of beam and columns.	10

Text Books:

1. Introduction to Linear Algebra (2nd edition) by Serge Lang, Springer

2. Elementary Linear Algebra (10th edition) by Howard Anton and Chris Rorres, John Wiley and sons.

3. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.

4. S. L. Ross, Differential Equations, 3rd Edition, Wiley India, 1984.

Reference Books:

1. Shanti Narayan, Differential Calculus, S. Chand & Co.

2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, New York.

3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi.

4. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II) , Pune Vidyarthi Griha Prakashan, Pune.

5. Differential Equations with Applications and Historical notes by George Simmons, Tata McGraw Hill publishing company Ltd, New Delhi

6. K.D Joshi, Calculus for Scientists and Engineers, CRC Press.

7. Prasad and Reena Garg, Advanced Engineering Mathematics, Khanna Publishing Company Private Limited, New Delhi.

8. Schaum's outlines of Linear Algebra (5th edition) by Seymour Lipschutz, Marc Lipson, McGraw-Hill Education (India) Private Limited, New Delhi

MGMUNIVERSITY

Course Code: APS21BSL104 **Course Name:** Engineering Chemistry **Course Category:**

Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, MSE-20, ESE–40

Pre-requisites: Fundamentals of basic chemistry.

Course Objectives:

1. The primary objective of an engineering chemistry course is to familiarize the students with new developments in engineering chemistry.
2. To impart a sound knowledge on the principles of chemistry involving the different application oriented topics required for all engineering branches.
3. The objectives of the laboratory sessions are to enable the learners to get hands-on experience on the principles discussed in theory sessions and to recognize the applications of these concepts in engineering.

Course Outcomes:

At the end of the course, students will be able to,

CO.1 : Illustrate the water quality parameters, water softening processes and causes of hard water in industries.

CO.2 : Demonstrate a comprehensive understanding of advanced concepts in polymer chemistry.

CO.3 : Apply fundamental concepts of corrosion science to solve problems arising in engineering applications.

CO.4 : Interpret physical, chemical properties and applications of fuels and lubricants.

CO.5 Describe the functions of batteries and applications of modern instrumental techniques like conductometry, pH metry, TLC, gas chromatography, UV-Visible and IR spectroscopy.

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Contents:

Unit	Content	Teaching Hours
1	**Water Treatment:** Natural sources of water, Impurities in water, Water quality parameters and its BIS standards, Hardness- Definition, Types, Estimation of hardness by EDTA method, Numerical based on hardness calculation, Disadvantages of hard water, Boiler troubles (causes, effect on boiler operation and methods of prevention), Internal and external treatments, Alkalinity and its determination, Water softening: Ion exchange process, Sewage water analysis- Dissolved oxygen (DO) and its determination, Biological oxygen demand (BOD), Chemical oxygen demand (COD) and their significance, Sewage water treatment.	09
2	**Polymer Chemistry:** Introduction, Classification of polymers, Use and disposal of polymers, Polymerization and its types, Plastics and its types- Thermoplastic and thermosetting plastics, Preparation, properties and engineering applications of: PVC, PMMA, Bakelite and Epoxy resin, Moulding constituents of	08

	plastics, Methods for moulding of plastics into articles, Conducting polymers and Biopolymers (Introduction, types, examples and its applications.	
3	**Corrosion and its Control:** Introduction, Types of corrosion, Mechanism of dry & wet corrosion, Factors influencing on corrosion – Nature of metal &Nature of environment. Methods of corrosion control, Cathodic and anodic protection, Use of Inhibitors, Protective Coatings: a) Metallic coatings: Types of coatings methods of applications, (hot dipping, cladding and electroplating), b) Nonmetallic coatings: Chemical conversion coatings, Powder coatings.	08
4	**Fuels and Lubricants:** **Fuels:** Introduction, Classification of fuel, Calorific value of a fuel, Characteristics of a good fuel, Solid fuel- Coal, Various types of Coal, Analysis of coal- Proximate and Ultimate analysis, Numerical based on analysis of coals, Liquid fuel- Refining of Petroleum, Gaseous fuels- LPG and CNG. **Lubricants**: Introduction, Mechanism of lubrication, Classification of lubricants, Solid, Semi-solid and Liquid Lubricants, Properties of lubricants, Physical properties – Viscosity & Viscosity index, Surface tension, Flash and Fire point, Cloud and pour point. Chemical properties – Acid value, Saponification value, Aniline point.	10
5	**Electrochemistry and Instrumental Methods of Chemical Analysis**: **Electrochemistry**: Introduction - Basic concepts: Conductance, Specific Conductance, Equivalent conductance, Molecular conductance, Effect of dilution on conductance, Cell constant. Battery: Primary & secondary batteries, Ni-Cd cell, Lithium-air battery, Fuel cell- H_2-O_2 Cell. **Instrumental Methods of Chemical Analysis**: Basic principle, instrumentation and applications of pH metry, Conductometry, Thin layer chromatography, Gas Chromatography, Ultraviolet-Visible spectroscopy and Infra-Red Spectroscopy	10

Text Books:

1. Jain P.C & Jain Monica, Engineering Chemistry, Dhanpat Rai & Sons, Delhi.
2. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New Delhi.
3. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
4. Shashi Chawla, Engineering Chemistry, Dhanpat Rai and Co Education and Technical Publishers.
5. Shikha Agrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers

Reference Books:

1. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
2. Atkins, Physical Chemistry, OUP Oxford.
3. Bhal & Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
4. V. R Gowarikar, Polymer Science, New Age International Ltd.
5. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Medi

E-Resources:

Name of the website/ E-Journals/ Online Videos

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/)
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-and-engineering/chemistry?languages=en&page=2)
4. "Introduction to Polymer Physics" NPTEL Course (https://www.youtube.com/playlist?list=PLwdnzlV3ogoXe67WsgE8f1fOIWcc5GKKS)
5. "Introduction to Corrosion" NPTEL Course (https://www.youtube.com/playlist?list=PL8lylDWRkaW8BXestE4baRnN0699S11Lq)
6. "Tribology & Lubrication" NPTEL Course (https://www.youtube.com/playlist?list=PLLwnvFq-JAltJvWAfEVU4gcUK27hEA7FD)
7. "Electrochemistry" NPTEL Course (https://www.youtube.com/playlist?list=PLVFqK_9GOGXnnriQpsn0z1Rss96Rh0vsm)
8. "Modern Instrumental Methods of Analysis" NPTEL Course (https://www.youtube.com/playlist?list=PL400CAFBA72E94CF8)

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Course Code: APS21ESL103 **Course Name:** Engineering Mechanics **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites:

1. Coordinate Geometry, Trigonometry, Sine & Cosine Rule, UnitConversions
2. Fundamentals of Physics

Course Objectives:

To develop capacity to predict the effect of force and motion in the course ofcarrying out the design functions of engineering.

Course Outcomes:

Students are able to,

CO1.Identify the force system for given conditions by applying the basics ofmechanics.

CO2. Determine the unknown forces of different engineering systems byapplying equilibrium conditions.

CO3. Apply the principles of friction and to locate Center of Gravity andfind Moment of Inertia of plane lamina.

CO4.Establish the relations between kinematic parameters for differenttypes of motion.

CO5.Formulate the relevant equations for types of motion in kinetics.

Contents:

Unit	Content	Teaching Hours
1	**Force System**: Introduction to Mechanics, Laws of mechanics, Newton's Laws, Law of Parallelogram, Law of transmissibility, Characteristics of force, System of Forces, Method of resolution and composition moment of a force, Law of Moments, Varignon's Theorem, Problems on moment, Resultant, Equivalent force& couple,properties of couple	08
2	**Equilibrium:** Introduction to Equilibrium and its types, Equilibrant, Concept of FBD,Analytical conditions of equilibrium, Equilibrium of different force system, Lami's Theorem, Types of loads, beams and supports.	07
3	**Friction, Centre of Gravity and Moment of Inertia:** Friction: Introduction to friction, types and application, Laws of friction, Angle of friction, Angle of repose, Cone of friction, Problems on horizontal & inclined plane, block, and ladder. CG & MI: Centroid of regular and composite plane lamina, MI and its application, Perpendicular axis Theorem, Parallel Axis Theorem, Radius of Gyration, Problems on plane and composite lamina.	08
4	**Kinematics:** Introduction and classification of dynamics, motion and its	04

	classification, Rectilinear Motion, Equation of Motion, Motion curves, Curvilinear Motion, rectangular and tangential components of acceleration, Projectile Motion: General Equation of Projectile Motion	
5	**Kinetics**: Basic concepts and laws of motion, D'Alemberts Principle, Problems on rectilinear motion, Curvilinear motion, Work Energy Principle, Work Done by force, Work Done by weight force, Work Done by frictional force, Work Done by spring force, Kinetic and Potential energy of the particle, Problems on all cases of Work Done, Principle of Impulse and Momentum, Principle of Conservation Momentum, Impact and its types, Coefficient of restitution, Problems on impact, Problems on impulse and momentum, Kinetics of rigid body problems.	03

Text Books:

1. Beer, F.P and Johnston Jr. E.R., "Vector Mechanics for Engineers (In SI Units): Statics and Dynamics", 8th Edition, Tata McGraw-Hill Publishing company, New Delhi (2004).
2. Vela Murali, "Engineering Mechanics", Oxford University Press (2010).
3. S. Timoshenko, D. H. Young, "Engineering Mechanics", McGraw Hill, 1995.
4. Tayal A. K., "Engineering Mechanics", Umesh Publications, 2010.
5. Singer F. L., "Engineering Mechanics - Statics & Dynamics", Harper and Row Pub. York.
6. Khurmi R. S., "Engineering Mechanics", S. Chand Publications, N. Delhi.

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Reference Books:

1. McLean, Nelson,"Engineering Mechanics",Schaum's outline Series,McGraw Hill Book Company, N. Delhi,Publication.
2. Hibbeller, R.C and Ashok Gupta, "Engineering Mechanics: Statics and Dynamics", 11thEdition,Pearson Education (2010).
3. Bhavikatti, S.S and Rajashekarappa, K.G., "Engineering Mechanics", New Age International (P)Limited Publishers, (1998).
4. Irving H. Shames and Krishna Mohana Rao. G., "Engineering Mechanics – Statics and Dynamics",4th Edition, Pearson Education (2006).
5. RajasekaranSandSankarasubramanianG.,"EngineeringMechanicsStaticsandDynamics", 3rdEdition,VikasPublishingHousePvt. Ltd., (2005).
6. Meriam J.L. and Kraige L.G., " Engineering Mechanics- Statics - Volume 1, Dynamics- Volume 2",Third Edition, John Wiley & Sons,(1993).
7. Kumar, K.L., "Engineering Mechanics", 3rd Revised Edition, Tata McGraw-Hill Publishing company, New Delhi (2008).

General Instructions:

E-Resources :

1. www.nptel.ac.in (Learning platform from IIT professors)
2. http://www.asnu.com.au (For Engineering applications)
3. www.discoveryforengineers.com (Investigating Discoveries)

Course Code: APS21ESL104 **Course Name:** Building Programming logic in C
Course Category:

Credits: 1 **Teaching scheme:** L-1 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Basic Knowledge of computers.

Course Objectives:

1. To introduce the basic concepts of programming and the C language, including algorithms, program structure, data types, variables, constants, and tokens.
2. To learn various operators, expressions, type casting, and input/output functions in C.
3. To understand decision-making statements, loops, and jump statements for controlling program flow.
4. To learn arrays, strings, and functions, including user-defined and library functions, recursion, and variable scope.
5. To understand structures, unions, and enums for creating user-defined data types in C

Course Outcomes:

After the completion of this course, students will be able to:

CO1: Write a simple C program.
CO2: Apply operators, expressions, and I/O functions to develop simple C programs.
CO3: Design and implement programs using control structures and loops for decision-making and iteration.
CO4: Write C programs using arrays, strings, and functions to solve problems efficiently.
CO5: Write C programs using structures, unions, and enums to organize and manage complex data efficiently.

Contents:

Unit	Content	Teaching Hours
1	**Introduction To C Programming:** **Overview of C**: History of C, Algorithm and Structure of C program. Keywords, Tokens, Data types, Constants, Literals and Variables. **Operators and Expressions:** Arithmetic operators, Relational operators, Logical operators, Expressions. Operator precedence and associativity, Type casting **Console I/O Formatting:**Formatted I/O, Unformatted I/O functions: getch(), getchar(), getche(), getc(), putc(), putchar()	03
2	**Control Statements and Loops:** **Control statements:** if , if-else, Nested if–else, conditional operators, switch and break, nested conditional branching statements, **Looping Statements** : while, do while, for, Nested loops, **Jump Statements**: break and continue, goto and label, exit function.	03
3	**Array and String:** **Array:** Array declaration, One and Two dimensional numeric and character arrays, Multidimensional arrays, operations on array. **String:** String declaration, initialization, string manipulation with library	03

	function, Without library functions.	
4	**Functions:** **Functions:** Definition, function components: Function arguments, return value, function prototype, function call statement, **Types of function:** Library Functions(math and character),User-defined Functions, Scope and lifetime of variable, Call by value and call by reference. Function using arrays, function with command line argument. **Recursive function.**	03
5	**Structure and Union:** **Structure:** Basics, declaring structure and structure variable, typedef statement, array of structure, array within structure, Nested structure; passing structure to function, function returning structure. **Union:** basics, declaring union and union variable, Difference between Structure and Union **Enum:** declaring enum and enum variable.	03

Text Books

1. Programming in ANSI C, E. Balagurusamy, Tata McGraw-Hill, Third Edition.
2. Let Us C, Yashwant Kanetkar, Infinity Science Press, Eighth Edition.
3. Mastering C, K. R. Venugopal, Tata McGraw-Hill.

Reference Books

1. The C Programming Language, Brian W. Kernighan & Dennis M. Ritchie, Prentice Hall, 2nd Edition.
2. Applications Programming in ANSI C, R. Johnsonbaugh & Martin Kalin, Macmillan, 2nd Edition.
3. The Spirit of C, Mullish Cooper, Jaico Publishing House.
4. How to Solve It by Computer, R. G. Dromey, Pearson Education

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Course Code: APS21PCL101 **Course Name:** Basics of Electrical and ElectronicsEngineering
Course Category:

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university mathematics.

Course Objectives:

1. Understand Electrical circuits and classify circuits asper laws.
2. Understand Magnetic circuits and apply them intransformer devices.
3. Understand the working principle of semiconductordevices.
4. Understand the basics and applications of digitalelectronics.

Course Outcomes:

After completion of this course, students will be able to:

1. Solve simple DC and single-phase AC circuits usingKCL, KVL and network theorems.
2. Explain fundamentals of magnetic circuit andtransformers.
3. Explain the working of diodes and transistors.
4. Explain the importance of number systems and logicgates.

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Contents:

Unit	Content	Teaching Hours
1	**Electrical Circuits:** Voltage and current sources: independent, dependent, ideal and practical; V-I relationships of resistor, inductor, mutual inductor and capacitor; Kirchhoff's laws, mesh and nodal analysis, superposition, Thevenin's, maximum power transfer theorems. Alternating voltages and currents, RMS, average, maximum values, Single Phase RL, RC, RLC series circuits, Power in AC circuits, Power Factor,Three phase balanced systems.	08
2	**Magnetic Circuits & Transformer:** Concepts of m.m.f, flux, flux density, reluctance, permeability and field strength, their units and relationship, right hand thumb rule and cork screw rule, Faraday's law of electromagnetic induction, Fleming's right-hand rule, statically and dynamically induced EMF's self and mutual inductance coefficient of coupling, energy stored in magnetic circuit, Single phase transformers: Construction, principle of working, e.m.f. equations.	08

3	**Semiconductor Devices:** Introduction to Semiconductors, P-type and N-type Semiconductors, P-N Junction Diode: Construction and working, V-I characteristics of Diode, Reverse breakdown mechanism. Special purpose diodes: Zener Diode, Light Emitting Diode (LED) and PhotoDiode - Construction, working and applications. Bipolar Junction Transistor (BJT): types, construction and regions of operation.	07
4	**Digital Electronics:** Binary, Decimal, Octal, Hexadecimal number systems and their inter-conversion, Binary Addition, Binary Subtraction, One's complement, Two's complement, Logic Gates: Basic, Universal and Special. Boolean Laws (AND, OR, NOT) and Demorgan's theorems, Realization of basic logic gates using universal gates.	07

TEXT BOOKS:

1. V. N. Mittal and Arvind Mittal, "Basic Electrical Engineering" McGraw Hill
2. Edward Hughes, "Electrical Technology,", Pearson Education
3. Electrical Technology Vol.1 & Vol.4 by B. L. Theraja & A. K. Theraja, S. Chand Publications
4. "Principles of Electronics", V.K. Mehta, S.Chand Publications.
5. "Electronics Devices and Circuits", S Salivahanan, McGraw Hill Publications.
6. "Modern Digital Electronics", R.P. Jain, McGraw Hill Publications.

REFERENCE BOOKS:

1. Vincent Del Toro, "Electrical engineering Fundamentals", PHI second edition 2011
2. Robert Boylestad, "Electronics Devices and Circuits Theory", Pearson Education India
3. "Electronics Devices and Circuits Theory", Robert Boylestad, Pearson Education India.

E-Resources:

1. Online course on NPTEL "Basic Electrical Engineering"
2. NPTEL Course on "Introduction to Basic Electronics", Prof. T.S. Natarajan, IIT Madras.
3. NPTEL Course on "Digital Electronic Circuits", Prof. Goutam Saha, IIT Kharagpur.

Course Code: APS21IKL1XX **Course Name:** Indian Knowledge Systems **Course Category:**

Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Nil

Course Objectives:
The objective of this course is
1. To make students understand foundational concepts in IKS for science, engineering and technology.
2. To explore ancient Indian pursuits and accomplishments in the various domains of engineering

Course Outcomes:
On completion of the course, the student should be able to:
CO 1: familiarise with key components of the IKS & develop appreciation for Indian philosophical systems.
CO 2: understand key features of Indian Numeral System, units of measurement and the framework for establishing the right knowledge.
CO 3:appreciate the unique & specific contributions of ancient Indian mathematicians in Arithmetic, Geometry & Trigonometry.
CO 4:develop awareness about engineering & technology heritage of India and understand ancient Indian contributions in various engineering domains.

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Contents:

Unit	Content	Teaching Hours
1	**Introduction to IKS:** Importance of ancient knowledge, defining IKS, Classification framework for IKS, Historicity of IKS, Indian philosophical systems, Vedic schools of philosophy(Sankhy and yoga, Nyaya and Vaisesika, Purva- Mimamsa and Vedanta), Non- Vedic philosophical systems (Jain, Bauddha and Carvaka) , Wisdom through the ages: Issues of interest in the Puranas, Itihasa as a source of wisdom (Uniqueness of the two epics), Nitisastras.	08
2	**Foundational Concepts for Science and Technology:** Number system in India, salient features of the Indian numeral system, Measurements for Time, Distance and weight, The knowledge triangle, Prameya- a Vaisesikan approach to Physical Reality, Pramana – The means of valid knowledge, Framework for establishing valid knowledge.	07
3	**Science in IKS:** Mathematics: Great mathematicians and their contributions, Arithmetic (square of a number, square root, series and progressions), Geometry (Property of right angled triangle in Sulba- sutras, value of π), Trigonometry, Algebra, Binary Mathematics and combinatorial problems in Chandah-Sastra of Pingala	07

4	**Engineering and Technology in IKS:** The Indian Science and Technology Heritage, Mining and ore extraction, Metals and metal working technology, Iron and Steel in India, Lost wax casting of idols and artefacts, apparatuses used for extraction of metallic components. Literary sources for Science and Technology, Physical Structures in India, Irrigation & Water Management, Dyes and Painting Technology, Shipbuilding.	08

Textbooks:

- Mahadevan B.,Bhat Vinayak Rajat & Nagendra Pavana R.N. "Introduction to Indian Knowledge System Concepts and Applications"PHI,2023.
- Jha Amit "Traditional Knowledge System in India" Atlantic Publishers and Distributors (P) Ltd,2023
- Chauhan Bhag Chand "IKS: The Knowledge system of Bharata" Garuda Prakashan,2023
- Bag A.K., "Mathematics in Ancient and Medieval India" Chaukhambha Orientalia, 1979
- Sengupta Nirmal,"Traditional Knowledge in Modern India"Springer,2019

Reference Books:

- Bag A.K., "Historyof Technology in India,Vol.I" Indian National Science Academy,1997
- Kumar Alok, "Ancient Hindu Science" Jaico Publishing House,2019
- Datta B and Singh A.N."History of Hindu Mathematics:Parts I and II" Asia Publishing House,1962
- Kapoor Kapil, Singh Awdhesh Kumar "Indian Knowlege Systems Vol.-I & II" D.K. Print World Ltd,2005

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E-Resources:

https://www.youtube.com/watch?v=-cBd6JYPWtY&list=PLRfu94TCePTtWtu0x145H_63WgoeYickE

https://www.youtube.com/watch?v=yvj5ROYbP2E&list=PLRfu94TCePTtLuEYSzmJXNYK_EnDSvY3N

https://iksindia.org/book-list.php

Course Code: APS21VSP102 **Course Name:** Workshop Practices **Course Category:**

Credits: 2 **Teaching scheme:** L-4 **Evaluation scheme:** CA–60, ESE–40

Pre-requisites: Pre-university English.

Course Objectives:

1.To know about the different carpentry tools and perform various carpentry operations to complete the job.

2. To understand different welding tools, joints, defects and perform welding operation to complete useful article/job.

3. To learn various types of pipes, plumbing tools, operations and perform thread cutting on GI pipes.

4. To know different sheet metal tools, operations, applications and perform various operations to complete job.

Course Outcomes:

On completion of the course, the student should be able to:

1. Perform basic carpentry operation on wood and prepare carpentry article.

2. Perform welding processes and prepare welding article by performing various welding operations.

3. Use plumbing tools, processes and perform threading on GI pipe.

4. Use sheet metal tools, processes and prepare sheet metal article.

Contents:

Unit	Content	Teaching Hours
1	**Carpentry Shop:** Carpentry shop: Types of woods, tools, joints, operations, applications, safety measures etc. Job: Exercises on wood involving operations marking, sawing, chiselling, planning, grooving etc to make useful wooden article/ job e.g. Wooden Trophies, Showpiece articles, Stools etc.	08
2	**Welding Shop:** Welding Shop: Types of welding, welding joints, tools, welding defects, applications, safety measures etc. Job: Exercise in Arc welding to make useful articles like Grills, Stools, Tree Guards, Flower pot stand, Shoe rack, Bag Stand, Showpiece Articles from Scrap etc.	07
3	**Plumbing Shop:** Plumbing shop: Study of types of pipe, pipe joints, operations, applications, safety measures etc. Job: Prepare threading on GI pipe to make useful items like Nipple, Pipe	07

	joints etc.	
4	**Sheet Metal Shop:** Sheet Metal Shop: Sheet metal tools, operations, applications, safety measures etc. Job: Making an utility item using G I sheet involving development, marking, cutting , bending, spot welding/riveting Parts like i) Tray, ii) Funnel etc.	08

- **Termwork:** Students shall maintain workshop diary which contains Job Drawing, information of tools, operations required to complete the jobs, records of job completions etc.
- **End Semester Exam:** ESE will be viva-voce based on jobs prepared by the students during the term.

Text Books:

2. Hazra and Chaudhary, Workshop Technology-I, Media promoters & Publisher private limited.

3. Hazra and Chaudhary, Workshop Technology-II, Media promoters & Publisher private limited.

Reference Books:

1. K. C. John, Mechanical Workshop Practice, Prentice Hall Publication, New Delhi, 2010.

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Course Code: APS21BSP102 **Course Name: Engineering Chemistry Lab Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

Course Outcomes:
Upon completion of the Lab Experiments, the students should able to,

LO.1 **:** Perform the experiments as well as accurately record the readings and analyze the results of such experiments.

LO.2 : Estimate the impurities present in water using titrimetric and instrumental methods.

LO.3 : Employ the basic techniques used in chemistry laboratory for analysis such as volumetric titrations, complexometric titrations, Conductometry, pH metry, viscometer, Stalagmometer and TLC

Contents:

Sr. No.	List of Practical	Lab Hours
1	Determination of hardness of water sample by EDTA method.	02
2	Determination of chloride content in water sample by precipitation titration method.	02
3	Determination of dissolved oxygen in water by Iodometric method.	02
4	Determination of percentage purity of bleaching powder.	02
5	Determination of strength of acid / base using pH metric titration.	02
6	Determination of strength of acid / base using conductometric titration.	02
7	To determine the cell constant of the given conductivity cell.	02
8	To determine relative surface tension of unknown liquids by using stalagmometer.	02
9	To determine the viscosity of unknown liquids by using Ostwald / Redwood viscometer.	02
10	To determine acidity of given water sample.	02
11	Determination of acid value of an oil sample.	02
12	Determination of saponification number of an oil sample.	02
13	To determine alkalinity of given water sample.	02
14	Preparation of phenol- formaldehyde / urea-formaldehyde resin.	02
15	To find out the Rf value of given sample by thin layer chromatography.	02
16	To separate the given mixture using thin layer chromatography.	02
17	Proximate analysis of coal (moisture content, volatile matter, ash content).	02

18	To study factors influencing on rate of electrochemical corrosion.	02
19	To determine flash and fire point of given oil by Pensky-Marten flash point apparatus.	02
20	To determine cloud and pour point of lubricating oil.	02

Reference Books:

1. A Text book on Experiments and Calculations in Engineering Chemistry by Dr. S. S. Dara, S Chand Publication.
2. Laboratory manual on Engineering Chemistry by S. K. Bhasin & Sudha Rani, Dhanpat Rai Publishing Company.
3. Engineering Chemistry with Laboratory Experiments by M. S. Kaurav, Asia-Pacific the holdings Private Ltd.

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Course Code: APS21ESP104 **Course Name:** Engineering Mechanics lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

Course Outcomes:
Upon completion of the Lab Experiments, the students should able to,
1.Describe the working principle of mechanics and correlate them with day to day engineering applications.
2.Formulate and solve mechanics problems based on law of moments, conditions of equilibrium.
3.Verify theoretical concepts through analytical, experimental and graphical methods.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Verification of law of Machine using Screw jack	02
2	Polygon law of coplanar forces.	02
3	Bell crank lever	02
4	Lami's theorem	02
5	Support reactions for beam	02
6	Problems on beam reaction by graphics statics method	02
7	Inclined plane (to determine coefficient of friction).	02
8	Centroid of irregular shaped bodies	02
9	Determine center of gravity for composite sections	02
10	Determine moment of inertia for composite sections	02
11	Moment of Inertia of fly wheel	02
12	Simple / compound pendulum	02

References:

1. "College of Engineering, Pune"
2. "University of Mumbai"

Course Code: APS21ESP105 **Course Name** : Building Programming logic in C Lab

Course Category:

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Basic Knowledge of computers.

Course Objectives:

1. To introduce the basic concepts of programming and the C language, including algorithms, program structure, data types, variables, constants, and tokens.
2. To learn various operators, expressions, type casting, and input/output functions in C.
3. To understand decision-making statements, loops, and jump statements for controlling program flow.
4. To learn arrays, strings, and functions, including user-defined and library functions, recursion, and variable scope.
5. To understand structures, unions, and enums for creating user-defined data types in C

Course Outcomes:

After the completion of this course, students will be able to:

CO1: Write a simple C program.
CO2: Apply operators, expressions, and I/O functions to develop simple C programs.
CO3: Design and implement programs using control structures and loops for decision-making and iteration.
CO4: Write C programs using arrays, strings, and functions to solve problems efficiently.
CO5: Write C programs using structures, unions, and enums to organize and manage complex data efficiently.

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Sr. No.	List of Practical	Lab Hours
1	a) Write a C program to find sum and average of three numbers. b) Write a C program to find the sum of individual digits of a given positive integer.	02
2	a) Write a C program to check whether the given number is perfect or not. b) Write a C program to check whether the given number is strong or not.	02
3	a) Write a C program to generate the first n terms of the Fibonacci sequence. b) Write a C program to generate prime numbers between 1 to n.	02
4	a) Write a C program to find the roots of a quadratic equation. b) Write a C program to perform arithmetic operations using switch statement.	02
5	a) Write a C program to find the largest and smallest number in a list of integers. b) Write a C program to sort the array in ascending order.	02
6	a) Write a C program to find whether the given matrix is symmetric or not. b) Write a C program to perform addition of two matrices.	02

7	a) Write a C program that displays the position or index in the main string S where the sub-string T begins, or -1 if S doesn't contain T. b) Write a C program to count the number of lines, words, and characters in a given text.	02
8	a) Write a C program to swap the values of two variables using: i) call by value ii) call by reference b) Write a C program using user-defined functions to determine whether the given string is palindrome or not. c) Write a C program using function to perform multiplication of two matrices.	02
9	a) Write a C program to use function to insert a sub-string into a given main string from a given position. b) Write a C program to find factorial of a given integer using recursive function. c) Write C program to find GCD of two integers by using recursive function	02
10	a) Write a C program to calculate total and percentage marks of a student using structure.	02

Text Books

4. Programming in ANSI C, E. Balagurusamy, Tata McGraw-Hill, Third Edition.
5. Let Us C, Yashwant Kanetkar, Infinity Science Press, Eighth Edition.
6. Mastering C, K. R. Venugopal, Tata McGraw-Hill.

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Reference Books

5. The C Programming Language, Brian W. Kernighan & Dennis M. Ritchie, Prentice Hall, 2nd Edition.
6. Applications Programming in ANSI C, R. Johnsonbaugh & Martin Kalin, Macmillan, 2nd Edition.
7. The Spirit of C, Mullish Cooper, Jaico Publishing House.
8. How to Solve It by Computer, R. G. Dromey, Pearson Education

Course Code: APS21PCP101 **Course Name:** Basics of Electrical and Electronics Engineering Lab **Course Category:**

Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:

1. Understand the DC circuit as per KCL & KVL and AC circuits as per theorems.
2. Understand the working of a single-phase transformer.
3. Understand the working of electronic instruments, components and logic gates.
4. Understand the working of a PN junction diode.

Course Outcomes:

After completion of this lab, students will be able to:

1. Use of KCL & KVL to solve DC circuits as well as use of AC theorems to solveAC circuits.
2. Demonstrate a single-phase transformer & its working.
3. Use electronic instruments, working of electronic components and logics gates.
4. Use of PN Junction Diode and its applications.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Verification of Loop Analysis and Nodal Analysis for DC Circuits.	02
2	Verification of Thevenin's Theorem for DC Circuits.	02
3	Verification of Maximum Power Transfer Theorem for DC Circuits.	02
4	Determination of Voltage, Current and Power Flow in Single Phase AC Circuitincluding R, L and C with Combination.	02
5	Determination of Magnetic Material Terms and EMF Induction.	02
6	Demonstration and Determination of Single-Phase Transformer Terms like Voltage Ratio and Turns Ratio.	02
7	Study of Electronic Instruments.	02
8	Study of Electronic Components.	02
9	Study of Logic Gates.	02
10	Study of V-I characteristics of a PN Junction Diode using V-Lab.	02
11	Study of Rectifier Circuits using Every Circuit simulation application.	02
12	Study of working and troubleshooting of Smartphone, Computer & TV (Case Study Approach).	02

Course Code: MGM82CCP104 **Course Name:** National Service Scheme (NSS) **Course Category:**

Credits: 2 **Teaching scheme:** P-2 **Evaluation scheme:**

Pre-requisites: Nil

Course Objectives: Nil

Course Outcomes: Nil

Contents:

Sr. No.	List of Practical	Lab Hours
1	**Introduction to National Service Scheme (NSS):** **Emergence of NSS in India (Historical Background) and its development.** Organizational Structure of National Service Scheme from National Level to College Level.Objectives of National Service Scheme (NSS) National Service Scheme (NSS) – Symbol and its meaning **Symbol of NSS and its meaning** Motto of National Service Scheme (NSS) Various prayers, inspirational songs to be used in NSS Programme.	08
2	**National Service Scheme (NSS) Regular Activities:** **Guidelines of Distribution of working hours or academic year.** **Classification of Regular Activities in the Society** • Rural • Urban • Campus • Need base with association **Associations in NSS Activities** • Govt. Organization • NGO Scope for Innovation (Self-Generated)	08
3	**Social Issues in India:** **Concept of Society, Community (Steps involved in evaluation of society)** • Features of Indian Society • Communities in India Basic Social Issues in India Family System, Division of labour, Cast System in India, Gender Issues, Regional Imbalance	07

4	**Indian Constitution and Social Justice:** **Indian Constitution** • Preamble • Structure • Features • Fundamental Rights & Duties **Social Justice** • Social Justice – the Concept and its features Contribution for Social Justice – Mahatma Jyotiba Phule, Dr. Babasaheb Ambedkar, Shahu Maharaj.	07

References:

1. National Service Scheme Manual (Revised) 2006, Government of India, Ministry of Youth Affairs and Sports, New Delhi.
2. University of Mumbai National Service Scheme Manual 2009.
3. Avhan Chancellor's Brigade-NSS Wing, Training camp on Disaster Preparedness Guidelines, March 2012.
4. Rashtriya Seva YojanaSankalpana- Prof. Dr. SankayChakane, Dr. Pramod Pabrekar, Diamond Publication, Pune.
5. National Service Scheme Manual for NSS District Coordinators, National Service Scheme Cell, Dept. of Higher and Technical Education, Mantralaya,
6. Annual report of National Service Scheme (NSS) published by Dept. of Higher and Technical Education, Mantralaya,
7. NSS Cell, Dept. of Higher and Technical Education, Mantralaya, UTKARSHA- Socio and cultural guidelines.

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Course Code: MGM73CCP203 **Course Name:** Fine Art **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The objective of this course is

- Students will be able to draw clean lines.
- Students will be able to explore shading from light to dark.
- Students will be able to draw perspective drawing.
- Students will be able to sketch landscape and portrait drawing.

Course Outcomes:
On completion of the course, the student should be able to:

- Students will be able to pay attention to core details in visualization.
- Students will be able to represent on paper what they have observed in terms of 3 and 2 dimensional objects and light and dark play of perspective.
- Students will be able to draw clean lines and neat figures which will gradually help them in fashion illustrations.

Contents:

Sr. No.	List of Practical	Lab Hours
1	**Introduction to Drawing:** • Explore pencil as medium for drawing and exploration of different types of grade of pencils. • Lines – Freeing hand movement by practicing various ways to draw lines (wavy/zigzag/diagonal/vertical/horizontal)	08
2	**3D objects with Shading:** • Exploring and sketching 3D objects with Shading/Hatching – Light to Dark (HB/2B/3B/4B/5B/6B) • Space Division – Studying of proportion, Perspectives- Coin / Box / Landscape.	07
3	**Sill Drawing:** • Introduction to landscape and portrait drawing. • Still drawing (perspective drawing of man- made object)	07
4	**Exploration of color mediums:** • Exploration of color mediums (chalk/ink/poster/pastels/water) • Landscape perspective of natural and man- made, charcoal rendering. • Still drawing of man- made and natural object by rendering and defining perspective.Ambedkar,Shahu Maharaj.	08

Reference Books:

- Mellem , Jeff; Sketching people & Live drawing basics.
- Belleville- Van Stone, France; Sketch: the Non-Artist's guide to inspiration technique & drawing life

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Course Code: MGM73CCP204 **Course Name:** Visual Art **Course Category:**

Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** CA–30, ESE–20

Pre-requisites: Nil

Course Objectives:
The objective of this course is
- To understand the actual work process in advertising market.
- To understand the growth and necessity of advertising in market.

Course Outcomes:
On completion of the course, the student should be able to:
- Students will be able to choose a topic for campaign design.
- Students will understand that how campaign design is necessary
- Student will understand the actual work process in advertising market.
- Student will understand the process of designing.

Contents:

Sr. No.	List of Practical	Lab Hours
1	Introduction to Campaign Design**:** • Topic for campaign design • It should be either product, service or social topic. • Mind mapping	08
2	**:** • Different types of media • New digital medias • Use of elective subject in campaign • Software use in designing	11
3	**Sill Drawing:** • Final layout • Printing in actual size media • Presentation on ppt of the topic includes artwork with rough work.	11

Reference Books:
- Advertising Campaign Design Just the Essentials - By Robyn Blakeman
- Campaign Strategies and Message Design: A Practitioner's Guide from Start to Finish - by Mary Moffitt (Author)
- Student should refer the previous knowledge about medias and refer the internet for help.

Suggested Tutorial / Practical:

Assignment I: Students will be required to submit mind map.

Assignment II: Students will be required to submit list of media selection.

Assignment III: Students will be required to submit different medias in digital form.

Assignment IV: Students will be required to submit different medias in print form.

Teaching Methods and learning activities-

- Lecture with power point projections
- Demonstration with practical
- Documentaries and Presentation
- Use of different medium and art form to make a new concept of work

Assessment Strategies-

Classroom lecture

Power point presentations

- **Group discussion**

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Syllabus of Semester-III

Course code: SCS24PCL201 **Course name:** Discrete Mathematics **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA –60, ESE–40
Pre-requisites: Pre-university mathematics
Course Objectives: The course aims for students to develop logical thinking and its application to computer science, especially to emphasize the importance of proving statements correctly and de-emphasize the hand-waving approach towards correctness of an argument. The subject enhances one's ability to reason and ability to present a coherent and mathematically accurate argument. The course shall cover proof systems, combinatorics, number theory and their relevance for the computer science.
Course Outcomes: At the end of the course, the students will be able to – **CO1**.Explain formal logic and different proof techniques. **CO2**.Recognize relation between different entities using sets, functions, and relations. **CO3.**Apply basic knowledge of counting &Probability distributions and their applications **CO4**.Learn advanced counting techniques, recurrence relations to analyze performance of algorithms **CO5**.Apply number theory concepts in cryptography.

Contents:

Unit	Content	Teaching Hours
1	**Set Theory, Logic and Proofs** The Foundations: Logic and Proofs:Propositional Logic, Applications of Propositional Logic, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference, Introduction to Proofs, Proof Methods and Strategy. Basic Structures: Sets, Functions, Sequences, Sums, and Matrices , Sets, set Operations, Functions, Sequences and Summations, Cardinality of Sets	8
2	**Relations, Functions, Recurrence Relations** Induction and Recursion: Mathematical Induction, Strong Induction and Well-Ordering, Recursive Definitions and Structural Induction, Recursive Algorithms, Program Correctness, Algorithms: Algorithms, The Growth of Functions, Complexity of Algorithms	8
3	**Counting** The Basics of Counting, Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients and Identities, Generalized Permutations and Combination, Generating Permutations and Combinations Discrete Probability, An Introduction to Discrete Probability, Probability Theory, Bayes' Theorem, Expected Value and Variance	8
4	**Advanced Counting Techniques** Applications of Recurrence Relations, Solving Linear Recurrence Relations, Divide-and-Conquer Algorithms and Recurrence Relations, Generating Functions, Inclusion–Exclusion, Applications of Inclusion–Exclusion	8

5	**Number & GraphTheory:** Fundamental concepts of number theory including divisibility, prime numbers, GCD and LCM, Euclidean and extended Euclidean algorithms, modular arithmetic, congruence relations, Euler's Totient function, and Fermat's Little Theorem. Types of graphs, degree of vertices, graph representations, paths and circuits, and connected graphs, Euler and Hamiltonian graphs, including Hamiltonian paths and cycles, necessary conditions, Dirac's and Ore's theorems, and applications like the Traveling Salesman Problem	8

Text books:

1. Discrete Mathematics and Its Applications by Rosen, Kenneth L. Mcgraw Hill - New Delhi 8th Edition
2. Discrete Mathematical Structures with Application to Computer Science by Tremblay, J.P. &Manohar, Mcgraw Hill - New Delhi 1997, 2011
3. Textbook of engineering mathematics By A.B. Mathur and V.P. Jaggi. 1999

Reference books:

1. Discrete Mathematical Structures For Computer Science By Kolman, B& Busby R.C,Prentice Hall Of India Pvt Ltd., New Delhi. Fifth Edition
2. Applied Discrete Structures for Computer Science by Alan Doerr &Kenneth, Galgotia Pub. Pvt.Ltd. New Delhi. 1989
3. Probability, Statistics and Random Process by T Veerarajan, TMH. 2008
4. An introduction to Probability theory and its Application By W Feller, Wiley. 1957

ICT/ MOOKS

1. https://nptel.ac.in/courses/111105035/32
2. https://nptel.ac.in/courses/111105035/1
3. https://nptel.ac.in/courses/102101056/11
4. https://nptel.ac.in/courses/111105041/8

Course code: SCS24PCL202 **Course name:** Data Structure **Course category:** PCC
Credits: 2 **Teaching scheme:** L- 2 Evaluation **scheme:** CA–60, ESE–40
Duration of External Exam: 2 Hrs.
Pre-requisites: Programming Language like C, C++
Course Objectives:
1. The course aims to teach programming and introduce elementary data structures.
2. To introduce the student to the concept of data structures through abstract data structures including lists, stacks, queues and graphs.
3. To assess how the choice of data structures and algorithm design methods affects the performance of programs
4. To choose the appropriate data structure and algorithm design method for a specified application.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Apply knowledge of basic programming constructs to design algorithms and programs
CO2: To use appropriate data structures like arrays, linked lists, stacks and queues to solve real-world problems efficiently.
CO3: To analyse different possible solutions to a program and select the most efficient one.
CO4: To impart familiarity with various sorting, searching and hashing techniques and their performance comparison.

Contents –

Unit	Content	Teaching hours
1	**Introduction -** Introduction to Data & Data Structures, types of Data Structure -Primitive & Non- Primitive, Concept of Abstract Data Type, Arrays, Types of Arrays, Memory representation of Array, traversing linear array, inserting and deleting elements in array. Structures, Functions, Recursive Functions, Introduction to time & space complexity. Searching – Linear Search and Binary Search. Sorting – Bubble sort , Selection sort , Insertion sort, Hashing: Hashing (Hash Function), Hash Collision, Collision resolution : Chaining Open Addressing, Linear Probing, Quadratic Probing, Double Hashing.	8
2	**Stack & Queues -** Stacks: Definition, Array Representation of stacks, Linked representation of stacks. Evaluation of a Prefix, Infix and Postfix Expression, Transforming Infix into Postfix expression. **Queues:** Definition, Array Representation of Queues, Linked representation of Queues, Types of Queues: Circular Queues, Double ended Queue, Priority Queue,Stack using two queues, Queue using two stack.	6

3	**Linked List & Trees:** Linked List: Representation of Linked Lists in memory, Insertion & deletion from Linked List, Traversing & Searching a Linked List, Types of Linked List : Circular Linked List and Doubly Linked Lists. **Trees–** Terminology, Binary Trees, Properties of Binary Trees, Array and liked representation of Binary Trees, Traversals-In order, Post order, Preorder, Binary Search trees- Definition, Insertion, Deletion, Traversal, and Searching. Evaluation of Expression, AVL Trees.	8
4	**Graph:** Definitions, Terminologies, Matrix and adjacency list representation of graphs. Traversal Methods: Breadth First Search and Depth First Search, Minimum Spanning Tree, Prim's & Krushal Algorithm.	8

Text Books: 1. "Data Structures with C", S. Lipschutz, Mc Graw Hill.2010

2. " Schaum's Outline of Data Structures", Lipschutz, S., McGraw-Hill

3. "Fundamentals of Data Structures in C++", E. Horowitz, Sartaj Sahni, D.Mehta, Universities Press.1995

Reference Books: 1. "Data Structures using C", E. Balaguruswamy, McGraw Hill 2013

2. "Data Structures using C" Aarron M. Tenenbaum, Yedidyah Langsam,Moshe J. Augensten , PHP

Online Resources: 1. https://archive.nptel.ac.in/courses/106/106/106106127/

2. https://nptel.ac.in/courses/106102064

Course code: SCS24PCL203 **Course Name**: Cyber Security and Cryptography Essentials **Course Category**: PCC
Credits: 2 **Teaching scheme**: L-2 **Evaluation scheme**: CA–40, ESE–60, TW–0, PR–0
Pre-requisites: Basic Computer Fundamentals and Networking
Course Objectives: 1. Understand basic cybersecurity concepts, threats, policies, and foundational security principles. 2. Understand and apply basic cryptographic techniques and secure communication methods. 3.Identify cyber threats and use basic security tools, VAPT techniques, and malware analysis methods. 4. Understand cyber laws, digital forensics, incident response, and business continuity practices.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Understand basic cybersecurity concepts, threats, security policies, networking, operating systems, and cloud security.
CO2: Understand and apply basic cryptography, secure communication protocols, and common web security concepts.
CO3: Identify cyber threats and use basic security tools, VAPT techniques, and malware analysis methods.
CO4: Understand cyber laws, perform basic digital forensics, handle incidents, and apply risk management and recovery practices

Unit	Content	Teaching hours
1	**Cybersecurity Fundamentals & Architecture** Introduction to Cybersecurity: CIA Triad (Confidentiality, Integrity, Availability), Security Threats, Vulnerabilities, and Risks (e.g., malware, phishing, ransomware). Security Policies and Models, Legal, Ethical, and Professional issues in cybersecurity. Networking and System Security Basics: TCP/IP and OSI Models fundamentals and their security implications. Operating System Security (Windows, Linux basics, hardening, access control). Cloud Security fundamentals (models, risks, and controls).	7

2	**Cryptography and Secure Communications** Cryptography: Classical Ciphers (Substitution, Transposition) Symmetric-Key Cryptography (e.g., DES, AES). Asymmetric- Key Cryptography (e.g., RSA, Diffie-Hellman). Hash Functions (e.g., MD5, SHA) and Digital Signatures,Network Security Protocols: Public Key Infrastructure (PKI) and Digital Certificates. Secure Protocols like SSL/TLS, IPSec, and VPNs. Wireless Network Security (e.g., WPA3),Application Security: Introduction to Web Application Security (e.g., OWASP Top 10). Common vulnerabilities like SQL Injection and Cross Site Scripting (XSS).	8
3	**Threat Management and Defensive Techniques** Security Controls and Monitoring: Firewalls (Types, Configuration). Intrusion Detection/Prevention Systems (IDS/IPS). Security Information and Event Management (SIEM). Access Control Mechanisms (Authentication, Authorization, Biometrics),Vulnerability Assessment and Penetration Testing (VAPT): Introduction to Ethical Hacking concepts and methodology. Scanning and Enumeration. Common exploitation techniques. Reporting and remediation planning,Malware Analysis and Reverse Engineering (Basic concepts).	7
4	**Cyber Law, Forensics, and Incident Response Cyber Law and Compliance** Overview of Information Technology (IT) Acts and other relevant cyber laws (e.g., GDPR, HIPAA - if globally focused). Intellectual Property and Data Privacy laws. Compliance frameworks and standards,Digital Forensics and Incident Response: Incident Response Lifecycle (Preparation, Detection & Analysis, Containment, Eradication & Recovery, Post-Incident Activity). Computer Forensics process	8

Text Books:

1) N. Godbole and S. Belapure, Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, New Delhi, India: Wiley India, 2011.
2) M. E. Whitman and H. J. Mattord, Principles of Information Security, 6th ed., Boston, MA, USA: Cengage Learning, 2017.
3) William Stallings – Cryptography and Network Security
4) Nelson, Phillips & Steuart – Guide to Computer Forensics and Investigations

Reference Books:

1) V. Nakta and E. Sood, Fundamentals of Cyber Crime & Cyber Safety: A Socio Legal Perspective, Satyam Law / IBP Books, 2021.

2) J. Graham, R. Howard and R. Olson, Cyber Security Essentials, CRC Press, 1st ed., 2011. .

Online Resources: 1. NPTEL / SWAYAM lectures.

Course Code: SCS24HSL204
Course Name: Business Management & Financial Accounting
Course Category: EEMC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Pre-requisites: Basic Management.
Course Objectives: 1. To Help the Students to Analyze the Risk and Return of Alternative Sources of Financing. 2. To Enable Students to Describe the Estimated Cash Flows from a Project, including Operating, Net Working Capital and Capital Spending. 3. To Provide the Students to Understand the related Information of Business Finance. Students can prepare Basic Financial Statements. 4. To Enable Students to Prepare Final Financial Statements.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Analyze the Risk and Return of Alternative Sources of Financing. **CO2:** Estimate Cash Flows from a Project, including Operating, Net Working Capital and Capital Spending. **CO3:** Define Basic Terminology used in Finance and Accounts. **CO4:** Prepare Financial Statements.

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Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to Business Management** 1.1 Aims, Objective and Function of Business Management, Principles of Management. 1.2 Concept of Business Finance, Goals & Objectives of Financial Management, Sources of Financing-Long Term Financing and Short Term Financing.	7
2.	**Business Capital Management** 2.1 Concept of Business Working Capital, Significance, Types. Adequacy of Working Capital, Factors Affecting Working Capital Needs. 2.2 Financing Approaches for Working Capital, Methods of Forecasting, Working Capital Requirements.	8
3.	**Basics of Financial Accounting** 3.1 Meaning, Scope and Importance of Financial Accounting, Financial Accounting - Concepts and Conventions, Classification of Accounts. 3.2 Rules and Principles Governing Double Entry Book-Keeping System, Meaning, Preparation of Journal, Ledger, Cash Book & Trial Balance.	7
4.	**Financial Statement Preparation and analysis** 4.1 Preparation of Financial Statements. Profit & Loss Account, Balance Sheet.	8

	4.2 Ratio Analysis.	

Text Books:

1. Financial Management by Khan & Jain, Text, Problem & Cases, Tata McGraw Hill Publication 5th Edition.
2. Tulsian Financial Management by Dr. P.C.Tulsian, S.Chand Publication 5th Edition.
3. Taxman's Financial Management by Ravi M. Kishore, Taxmann 2017 Edition.

Reference Books:

1. A Textbook of Financial, Cost & Management Accounting by Dr. P.Periasamy, Himalaya Publishing House.
2. Fundamentals of Financial Management by Bhabhtosh Banerjee, PHI publication, 2nd Edition.

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Course Code: MGM56VEL102
Course Name: Constitution of India
Course Category: VEC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:30 and ESE:20
Pre-requisites: None.
Course Objectives: 1. Understand the Constitution and its importance. 2. Sensitize the students about Fundamental Rights and Duties Enshrined under Indian Constitution. 3. Familiarize students with the working of Indian Constitution.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Understand and Explain the Significance of Indian Constitution as the Fundamental Law of the Land. **CO2:** Exercise Fundamental Rights in Proper Sense at the Same Time Identify Responsibilities in National Building. **CO3:** Analyze the Indian Political System, Electoral Process, Emergency Provisions and Amendment Procedure. **CO4:** Understand the Powers and Functions of the Union, State and Local Governments in detail.

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Contents:

Unit No.	Content	Teaching Hours
1.	**Historical Background and Philosophy of Indian Constitution** 1.1 Societies before and after the Constitution Adoption, Introduction to the Indian constitution, Making of the Constitution. 1.2 Role of the Constituent Assembly, Preamble and Nature of Indian Constitution, Salient Features of Indian Constitution.	8
2.	**Fundamental Rights & Duties** 2.1 Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion. 2.2 Cultural and Educational Rights, Right to Constitutional Remedies, Fundamental Duties.	8
3.	**Directive Principles of State Policy and Other Important Constitutional Provisions** 3.1 Meaning and Significance of Directive Principles, Economic Principles, Political Principles. 3.2 Emergency Provisions under Indian Constitution, Election Commission.	8
4.	**Organs of Government** 4.1 Centre and State Legislature, Centre and State Executive, Centre and State Judiciary, Local Self Government.	6

Text Books:

1. D. D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2013.
2. Pravin Kumar, Basic Mechanical Engineering, Pearson Education; Second Edition, 2018.
3. M. P. Jain, Outline of Indian Legal and Constitutional History, Lexis Nexis, 2014.
4. Constitutional Law of India - J.N.Pandey.

Reference Books:

1. Austin, The Constitution of India, Cornerstone of a Nation. Oxford University press 1966.

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Course code: SCS24PCP202 **Course name:** Data Structures Lab **Course category:** PCC
Credits: 1 **Teaching scheme:** P- 2 **Evaluation scheme:** TW-30, PR–20
Pre-requisites: Programming Language such as: C, C++
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Implement programs using arrays and structures to solve computational problems.
LO2: Design and demonstrate searching, sorting, and hashing algorithms using appropriate data structures.
LO3: Implement and test stack and queue operations using both arrays and linked list representations.
LO4: Develop programs to create, manipulate, and demonstrate various linked list structures and their operations.
LO5: Implement and evaluate different tree data structures such as binary trees, binary search trees.
LO6: Develop programs to represent and analyze graphs using adjacency matrix and adjacency list, and design efficient graph algorithms.

List of Experiments–

Sr. No.	Name of Experiment	Practical Hours
1	Write a ‘C’ program to perform following Operations on Array: Create, Insert, Delete,Display.	2
2	Write a ‘C’ Program to Search a particular data from the given Array of Strings using Linear Search Method.	2
3	Write a ‘C’ program to Search a particular data from the given Array of numbers using Binary Search Method.	2
4	Write a ‘C’ Program to Sort an Array of numbers using Bubble Sort Method.	2
5	Write a ‘C’ Program to Sort an Array of numbers using Selection Sort Method	2
6	Write a ‘C’ Program to Sort an Array of numbers using Insertion Sort Method.	2
7	Write a 'C' Program to perform PUSH and POP Operations on Stack using an Array.	2
8	Write a 'C' Program to perform INSERT and DELETE Operations on Linear Queue using an Array.	2
9	Write a 'C' Program to perform INSERT and DELETE operations on Circular Queue using an Array.	2
10	Write a 'C' Program to Implement Singly Linked List with Operations: (i) Insert at beginning, (ii) Search, (iii) Display	2
11	Write a 'C' Program to Implement BST (Binary Search Tree) and Traverse in In-Order.	2

12	Write a program to implement Graph operations.	2

Text Books: 1. "Data Structures with C", S. Lipschutz, Mc Graw Hill.2010
2. " Schaum's Outline of Data Structures", Lipschutz, S., McGraw-Hill
3. "Fundamentals of Data Structures in C++", E. Horowitz, Sartaj Sahni, D.Mehta, Universities Press.1995
4. "Data Structures using C", E. Balaguruswamy, McGraw Hill 2013
Reference Books: 1. "Data Structures using C" Aarron M. Tenenbaum, Yedidyah Langsam,Moshe J. Augensten , PHP
Online Resources: 1. https://archive.nptel.ac.in/courses/106/106/106106127/
2. https://nptel.ac.in/courses/106102064

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Course code: SCS24PCP203 **Course name:** CodeLock: Cybersecurity & Cryptography Essentials Lab **Course category:** PCC
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation scheme:** CA–0, MSE–0, ESE–0, TW–30, PR–20

Pre-requisites: Discrete Mathematics, Basic Programming, Computer Networks
Course Objectives: 1.Introduce the fundamental principles of cryptography and its role in cyber security. 2.Explain classical and modern cryptographic algorithms, including block and stream ciphers. 3.Develop mathematical foundations for cryptography using number theory and algebra. 4.Describe public-key cryptographic algorithms and their applications. 5.Analyze authentication mechanisms, hash functions, and digital signatures. 6.Examine modern cryptographic systems, attacks, and emerging security technologies.
Laboratory Outcomes: At the end of the course, the students will be able to -
LO1: Explain the fundamental concepts and objectives of cryptography and classical cryptosystems.
LO2: Apply mathematical concepts like modular arithmetic and number theory to design cryptographic algorithms.
LO3: Analyze and compare symmetric and asymmetric cryptographic algorithms.
LO4: Demonstrate the use of digital signatures, authentication, and key management techniques.
LO5: Evaluate modern cryptographic systems and analyze cryptanalytic attacks and countermeasures.

List of Experiments–

Sr. No.	Name of Experiment	Practical Hours
1.	Implementation of Classical Ciphers (Caesar, Monoalphabetic)	2
2.	Implementation of Transposition Techniques (Rail Fence and Columnar Transposition)	2
3.	Simulation of Feistel Structure (Mini Block Cipher Model)	2
4.	Implementation of DES and Triple DES Algorithms	2
5.	Demonstration of Block Cipher Modes of Operation (ECB, CBC, CFB, OFB, CTR)	2
6.	Implementation of LFSR-based Stream Cipher	2
7.	Implementation of Modular Arithmetic and Extended Euclidean Algorithm	2
8.	Implementation of RSA and Diffie–Hellman Key Exchange	2

9.	Implementation of AES Algorithm	2
10.	Implementation of Hash Functions and Digital Signatures (SHA-256 / MD5 / DSS)	2

Text Books:

1. William Stallings, *Cryptography and Network Security: Principles and Practice*, Pearson Education.

2. Behrouz A. Forouzan, *Cryptography and Network Security*, McGraw Hill.

Reference Books:

1. Alfred J. Menezes, Paul C. van Oorschot, Scott A. Vanstone, Handbook of Applied Cryptography.
2. Jonathan Katz and Yehuda Lindell, Introduction to Modern Cryptography.

Online Resources: 1. NPTEL / SWAYAM lectures.

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Course code: SCS24FPJ205 **Course name:** Web Technology Lab **Course category:** FP **Credits:** 2 **Teaching scheme:**P-4 **Evaluation scheme:** TW : 30, PR:–20
Pre-requisites: Basic Concepts of Web
Course Objectives:
1. To get familiar with client-server architecture.
2. To explore recent web technologies.
Laboratory Outcomes: At the end of the course, the students will be able to -
LO1: Design web pages using HTML , CSS and JavaScript.
LO2: Build appropriate server-side scripts using PHP.
LO3: Create dynamic web application using PHP and MySQL.
LO4: Develop solution to real world problem using appropriate web technologies and content management system.

Sr. No.	List of Experiments	Practical Hours
1	Develop informative web pages using HTML5 (Use all possible formatting for example font, colour etc.). Web page should include various images, links within the page, links to other pages for navigation, new tabs.	02
2	Design an online registration form for any application.	02
3	Use Cascaded Style Sheets(CSS) to style the web pages designed. Make use of border, margins, padding, navigation, dropdown list etc.	02
4	Design an XML document to store information about a student in an engineering college affiliated to MGM University. The information must include RegistrationNo, Name, Name of Brach, Year of Joining, and e- mail id. Make up sample data for 3 students. Create a CSS style sheet and use it to display the document.	02
5	Design, Develop and Deploy new web application using Bootstrap in consultation of course instructor.	02
6	Write a Java Script to design a simple calculator to perform the following operations: sum, product, difference and division.	02
7	Write a JavaScript that calculates the squares and cubes of the numbers from 0 to 10 and outputs HTML text that displays the resulting values in an HTML table format.	02
8	Develop and demonstrate a HTML5 file that includes JavaScripts that uses functions for the following problems: Parameter: A string Output: The position in the string of the left-most vowel. Parameter: A number Output: The number with its digits in the reverse order.	02
9	Write a JavaScript code that displays text “TEXT- GROWING” with	02

	increasing font size in the interval of 100ms in RED COLOR, when the font size reaches 50pt it displays "TEXT-SHRINKING" in BLUE colour. Then the font size decreases to 5pt	
10	Write a PHP program to keep track of the number of visitors visiting the web page and to display this count of visitors, with proper headings.	02
11	Write a PHP program to display a digital clock which displays The current time of the server.	02
12	Write a PHP program to display the student records which are stored in the database.	02
13	Write the PHP programs to do the following: a. Implement simple calculator operations. b. Find the transpose of a matrix. c. Multiplication of two matrices. d. Addition of two matrices.	02
14	Develop web site using any CMS tool which falls into one of the categories blogs, social networking, News updates, Wikipedia, E commerce store. Website must include home page, and at least 3 pages.	02
15	Study Experiment/Project: Develop a web application project using the languages and concepts learnt in the exercises with a good looks. You can use any web technologies and Frameworks and databases.	02

Text Books: 1. Ralph Moseley & M.T. Savaliya,—Developing Web Applications ‖, Wiley publications, 2007

2. Kogent Learning Solutions Inc,Web Technologies, Black book, Dreamtech Press.2009

Reference Books: 1. IvanBayross, Web Enabled Commercial Application Development Using HTML,
JavaScript, DHTML and PHP, BPB Publications.2005, Third Edition

2. StevenM. Schafer," HTML, XHTML and CSS", Wiley India Edition., 5th Edition 2011

Online Resources: 1. MicrosoftAcademyCoursesonHTML5&CSS.3
2. Udemy and Coursera Courses on Web Technologies.
3. Spoken Tutorial Course on PHP and MySql.
4. W3Schools,Tutorials Point tutorials on HTML,CSS,JavaScript,PHP,Bootstrap.

Syllabus of Semester-IV

Course Code: SCS24PCL251 **Course Name:** Ethical Hacking **Course Category:** PCC
Credits: 2 **Teaching scheme**: L-2 **Evaluation Scheme**: CA: 60, ESE: 40
Duration of External Exam: 2 Hrs
Pre-requisites: Networking, Programming, Cyber security Basics.
Course Objectives:
1. Understand the fundamentals of ethical hacking, information security concepts, and legal and ethical considerations.
2. Analyze footprinting, reconnaissance, and network scanning techniques along with their countermeasures.
3. Explain enumeration, sniffing, and password cracking methods and evaluate their prevention techniques.
4. Identify malware, network attacks, social engineering techniques, and appropriate security defences.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Understand ethical hacking concepts, legal issues, and information security models.
CO2: Apply foot printing, reconnaissance, and network scanning techniques with countermeasures.
CO3: Make use of the tools that can be used to perform information gathering and apply foot printing countermeasures.
CO4: Understand about Enumeration, Sniffing and apply countermeasures against the Password Attacks.
CO 5: Apply countermeasures against the malicious program.

Contents –

Unit	Content	Teaching hours
1	**Introduction to Ethical Hacking** Introduction to Ethical Hacking, hacker classifications and their roles, legal and ethical considerations in hacking, techniques of hacking, and an introduction to information security. Information security models including computer security, network security, service security, application security,	8

	and security architecture. Information security program and its processes, component parts of an information security program, risk analysis in ethical hacking, and an overview of active and passive threats and exploits.	
2	**Foot printing, Reconnaissance and Scanning networks** Footprinting concepts including active and passive footprinting, purpose and objectives of footprinting, footprinting threats, types of footprinting, and footprinting countermeasures. Network scanning and its types, objectives of network scanning, scanning live systems, scanning techniques such as TCP connect or full open scan, types of stealth scans, and port scanning countermeasures. IDS evasion techniques, banner grabbing and its tools, vulnerability scanning, proxy servers and anonymizers, IP spoofing, and its countermeasures.	8
3	**Enumeration and sniffing, Password Cracking** Enumeration and sniffing concepts including definition of enumeration, enumeration techniques and types, and enumeration countermeasures. Sniffing concepts such as wiretapping and its types, packet sniffing, sniffing threats, working of sniffers, and sniffing methods including ARP spoofing and MAC flooding. Active and passive sniffing, types of sniffing attacks, sniffing countermeasures, and sniffing detection techniques. Password hashes and password cracking types including dictionary and brute-force attacks, password cracking using John the Ripper and other tools, password storage mechanisms in Linux (/etc/passwd and /etc/shadow) and Windows, testing and hardening of SSH passwords, and password cracking countermeasures.	8
4	**Trojans and other Attacks** Worms, viruses, Trojans, and types of worms and viruses, prevention of malware attacks, and various types of attacks such as DoS and DDoS. Waterhole attacks, brute force attacks, phishing and fake WAP, eavesdropping, man-in-the-middle attacks, buffer overflow, DNS poisoning, ARP poisoning, and identity theft. IoT attacks, bots and botnets, steganography in text, image, audio, and video. Types of social engineering including physical, remote, and hybrid social engineering	8

Text Books:

1. EC-Council "Ethical Hacking and Countermeasures Attack Phases", Cengage Learning
2. James S. Tiller, "The Ethical Hack: A Framework for Business Value Penetration Testing", Auerbach Publications, CRC Press.
3. The Basics of Hacking and Penetration Testing, Patrick Engebretson 2nd edition Syngress.
4. Ethical Hacking A Comprehensive Beginner's Guide to Learn and Master Ethical Hacking, By Hilary Morrison, hein smith · 2018, Create Space Independent Publishing

Platform
Reference Books: **1**. EC-Council, "Ethical Hacking and Countermeasures Attack Phases", Cengage Learning. **2.** Michael Simpson, Kent Backman, James Corley, "Hands-On Ethical Hacking and Network Defense", Cengage Learning. **3.** Matt Walker, All-In-One-CEH-Certified-Ethical-Hacker-Exam Guide

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Course code: SCS24PCL252 **Course name:** Operating System **Course category:** PCC
Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40
Duration of External Exam: 2 Hrs.
Pre-requisites: Computer Organization and Architecture & Fundamentals of Data Structures.
Course Objectives:
1. Provide basic knowledge of computer operating system structures and functioning.
2. Compare several different approaches to memory management, file management and process management
Course Outcomes: At the end of the course, the students will be able to -
CO1: Outline the basic concept of operating systems
CO2: Examine the working of various scheduling/allocation approaches
CO3: Understand deadlock, prevention and avoidance algorithms.
CO4: Compare and contrast various memory management schemes.
CO5: Describe file systems from disk management to high-level operations

Contents –

Unit	Content	Teaching hours
1	**Introduction –** Operating System overview objectives and functions, Types of Operating system, The Evolution of Operating System, Operating System Structure and Operations- System Services, Systems Calls, System Programs, System structure	6
2	**Process & CPU Scheduling -** Process Concept, Process Scheduling, and Operations on Processes, Inter process Communication, Threads, Interprocess Communication, Scheduling criteria, Scheduling algorithms and performance evaluation.	6
3	**Process Synchronization -** Critical Section Problem, Mutex Locks, Classical Problems of synchronization, Semaphores, Monitors, Deadlock- Principles of Deadlock, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.	8
4	**Memory Management & File Management-** Basic concept, Logical and Physical address map, Memory Partitioning: Fixed Partitioning, Dynamic Partitioning, Paging, Segmentation, Virtual Memory - Basics of Virtual Memory – Page Replacement Algorithms: Optimal, First in First Out (FIFO), Not recently used (NRU) and Least Recently used (LRU). **File Management:** Overview, File Organization and Access, File Directories, File System structure, Allocation methods (contiguous, linked,	10

	indexed), directory implementation efficiency and performance	

Text Books:
1. Abraham Silberschatz, Peter B. Galvin and Greg Gagne, Operating System Concepts, Wiley Publication, 8th Edition, 2008.
2. William Stallings, Operating System: Internals and Design Principles, Prentice Hall, 8thEdition.
3. Andrew S. Tanenbaum, Modern Operating System, PHI Publication, 4th Edition, 2015.
Reference Books:
1. D. M. Dhamdhere, Systems Programming and Operating Systems, McGraw-Hill,2ndEdition, 1996.
2. Garry Nutt, Operating Systems Concepts, Pearson Publication, 3rd Edition, 2003.
3. Harvey M. Deitel, An Introduction to Operating Systems, Addison-Wesley Publication, 2 ndEdition,1990
4. Tom Adelstein and Bill Lubanovic, Linux System Administration, O'ReillyMedia

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Course code: SCS24PCL253 **Course name:** Object Oriented Programming Using C++ **Course category:** PCC
Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40

Duration of External Exam: 2 Hrs.
Pre-requisites: Basics of programming concept , Basics of DS, Fundamental of Procedure Oriented
Course Objectives:
1. To develop a foundational understanding of object-oriented programming concepts using C++
2. To develop the understanding for designing and implementation of classes and objects
3. To develop the ability to understand reusability concept using inheritance in OOP
4. To develop the ability to handle runtime errors using exception handling techniques
Course Outcomes: At the end of the course, the students will be able to -

CO1: understand and apply object-oriented programming concepts using C++ fundamentals
CO2: design, implement, and manipulate classes and objects
CO3: analyse, implement inheritance to design reusable and hierarchical object-oriented programs**.**
CO4: apply exception handling concept to detect, manage, and recover from runtime errors effectively

Contents –

Unit	Content	Teaching hours
1	**Introduction to C++:** Introduction to C++, Layout of C++ program, Data types, variables, constants, Keywords, Operators. **Control statements:** Branching Statements: if, if-else, nested if, Break, continue and switch statement. Looping Statements: While, Do-while and for Statement.	6
2	**Classes , Objects and Templates:** Features of Object Oriented Programming, Class specification, Access Specifies, Defining Member Functions, Objects Declaration, Accessing Data Members and Member Functions, Constructors, Destructor, Friend Functions. **Templates:** Function Templates, Class Templates	8
3	**Inheritance and Polymorphism:** Introduction to inheritance, Base Class,	8

	Derived class, Types of Inheritance. **Polymorphism:** Function Overloading, Function Overriding, Virtual Functions, Operator Overloading.	
4	**Exception Handling:** Introduction, Exception Handling Mechanism, Handling Multiple Exceptions.	8

Text Books: E. Balagurusamy , Object Oriented Programming with C++ , MC Graw Hill Publication
Reference Books: 1. Tony Gaddis, Starting out with C++: from control structures through objects (7e) 2. B. Stroupstrup, The C++ Programming Language, Addison Wesley, 2004. 3. Problem Solving with C++ by Walter Savitch, Addison Wesley
Online Resources: NPTEL

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Course Code: SCS24HSL254
Course Name: Entrepreneurship Development
Course Category: EEMC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Pre-requisites: Knowledge of Planning.
Course Objectives: 1. Describe the Various Functional Areas of Management. 2. Essentials to Avoid Failure in Entrepreneurship.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** To Develop Skills related to Various Functional Areas of Management (Marketing Management, Financial Management, Operations Management, Personnel Management etc.). **CO2:** To Develop Skills related to Project Planning And Business Plan Development. **CO3:** To Determine and Compare Various Skills of Successful Entrepreneurs, Business Organizations and Business Laws. **CO4:** To Examine Essentials to Avoid Failure in Entrepreneurship.

Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to Entrepreneurship Development** Entrepreneurship Development, Evolution of Entrepreneurship, Meaning, Definition and Concept of Enterprise,Concepts of Entrepreneurship, Role of Entrepreneurship in Economic Development, Factors Affecting Entrepreneurship,Entrepreneurial Development models and Theories, Entrepreneurial Trait Tests, Business Opportunity Identification.	08
2.	**Entrepreneur** Major Types of Entrepreneurship, Techno Entrepreneurship, Women Entrepreneurship, Social Entrepreneurship, Intrapreneurship (Corporate entrepreneurship), Rural Entrepreneurship, Family Business etc. ,The Entrepreneurial Decision Process, Entrepreneurial Environment – Political, Legal, Technological, Natural, Economic, Socio – Cultural etc.	07
3.	**Project Planning, Skills for Successful Entrepreneurs & Business Organizations and Business Laws:** Project Report, Project Appraisal, Setting Up an Industrial Unit – Procedure And Formalities in Setting Up an Industrial Unit; Business Plan Development, Communication Skills, Creativity and Problem-Solving, Innovation, Negotiation Skills, Risk Management,Types of Business Organizations – Sole, Proprietorship, Joint Hindu Family Business, Partnership, Limited Liability Partnership (LLP), Corporate Governance, Franchising, Business Laws in India to Start Business.	08

4.	**Concepts of Entrepreneurship Failure** Issues of Entrepreneurial failure, Reasons of Entrepreneurial Failure, Essentials to Avoid Failure in Entrepreneurship, Case Study of Failure in Entrepreneurship.	07

Text Books:

1. Dynamics of Entrepreneurship Development by Vasant Desai.
2. Entrepreneurship Development and small business management by Poornima M. Charantimath.
3. Dynamics of Entrepreneurship Development by Vasant Desai.
4. Innovation and Entrepreneurship by Peter F. Drucker
5. Kathleen R Allen, Launching New Ventures, An Entrepreneurial Approach, Cengage Learning, 2016.
6. Vijay Sathe, Corporate Entrepreneurship, Cambridge, 2009.

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Course code: MGM21VEL101 **Course name:** Environment Studies **Course category:** VEC
Credits: 2 **Teaching scheme:** L-2 hrs/week **Evaluation scheme:** CA–30, ESE–20
Duration of ESE: 01 Hour
Pre-requisites: Basic Science
Course Objectives:
1. To study the environment and ecosystems.
2. To study different types of natural resources
3. To study concept of biodiversity and its conservation
4. To study the concept of causes, effects and control of different types of environmental pollution
5. To study population growth and its impact on environment
Course Outcomes: At the completion of the course, students should be able to:
1. Describe environment and ecosystem
2. Explain environmental impacts of human activities on natural resource
3. Identify the importance of conservation of biodiversity.
4. Describe the problems of environmental pollution, its impact on human and ecosystem and control measures
5. Identify the impact of increased population on environment

Contents –

Unit	Content	Teaching hours
1	**Ecosystem and the Environment** Definition, Scope and importance of Environmental studies, Need for Public awareness, Concepts of an Ecosystem, Structure and functions of an Ecosystem, Producers, Consumers and Decomposers in an Ecosystem	06
2	**Natural Resources** Renewable and non-renewable resources, Role of individuals in conservation of natural resources for sustainable lifestyles,Use and over exploitation of Forest resources, Deforestation, Timber extraction, Mining, Dams and their effects on forest and tribal people,Use and over exploitation of surface and ground water resources, Floods, Drought, Conflicts over water, Dams- benefits and problems.	06
3	**Biodiversity and its Conservation** Definition, Genetic, Species and Ecosystem diversity, Bio-geographical classification of India,Value of biodiversity: Consumptive use, Productive use, Social, Ethical, Aesthetic and option values.	06
4	**Environmental pollution** Definition, Causes, effects and control measures of Air pollution. Water pollution and Soil pollution,Causes, effects and control measures of Marine	06

	pollution, Noise pollution, Thermal pollution, Nuclear hazards	
5	**Population issues and the Environment** Population growth, Variation among nations, Population explosion- Family welfare programmes, Environment and Human health, Human rights, Value education, HIV/AIDS, Woman and Child welfare, Role of information technology in environment and human health, Case studies	06

Text Books:

1 BharuchaErach (ed) Text Book of Environmental Studies., University Press (India) Pvt. Ltd
2. BharuchaErach, 2003. The Biodiversity of India, Mapin Publishing Pvt. Ltd, Ahmedabad – 380013
3. Kaushik, Anubha& Kaushik, C.P. 2006. Perspectives in Environmental Studies, New Age International (P) Ltd. Publisher, New Delhi
4.Deswal, S. and Deswal, A. 2005. A Basic Course in Environmental Studies. Dhanpat Rai & Co ltd., Delhi.

Reference Books:

1. IndiaSantra S.C, (ed), Environmental Science, New Central Agency Pvt Ltd. Kolkata, India.
2. Botbin, D., and Keller, E.1995. Environmental Science. John Wiley and Sons, USA.
3. Cunningham, W.P. and Saigo, B.W. 1997. Environment Science. McGraw-Hill, USA.
4. Lal, R. and Stewart, B.A. 1992. Advances in Science: Soil Restoration Springer Verlag, New York.

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Course Code: SCS24PCP251 **Course Name:** Ethical Hacking Lab **Course Category:** PCC
Credits: 2 **Teaching scheme**: P-2 **Evaluation Scheme**: TW:50
Duration of External Exam: 2 Hrs
Pre-requisites: Networking, Programming, Cyber security Basics.
Course Objectives:
1. Understand the fundamentals of ethical hacking, information security concepts, and legal and ethical considerations.
2. Analyze footprinting, reconnaissance, and network scanning techniques along with their countermeasures.
3. Explain enumeration, sniffing, and password cracking methods and evaluate their prevention techniques.
4. Identify malware, network attacks, social engineering techniques, and appropriate security defences.
Lab Outcomes: At the end of the course, the students will be able to -
LO1: Understand how to gather and review information related using different foot printing techniques.
LO2: Understand the different types of password attacks and assess the password hashes and Password strength.
LO3: Understand Types of Virus, Trojans, Backdoor, and Key logger.
LO4: Understand website information like archived website and extract contents of a website.

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Foot printing and Reconnaissance: Performing foot printing using Google Hacking, website information, information about an archived website, to extract contents of a website, to trace any received email, to fetch DNS information.	2
2.	Scanning networks, Enumeration and sniffing: Use port scanning, Network scanning tools, IDS tool, sniffing tool and generate reports.	2

3.	Malware Threats: Worms, viruses, Trojans: Use Password cracking, Dictionary attack, Encrypt and decrypt passwords, DoS attack, ARP poisoning in windows, ipconfig, ping, net stat, trace route, Steganography tools.	6
4.	Developing and implementing malwares: Creating a simple key logger in python, creating a virus, creating a Trojan.	6
5.	Hacking web servers, web applications: Hacking a website by Remote File Inclusion, Disguise as Google Bot to view hidden content of a website, to use Kaspersky for Lifetime without Patch	2
6.	SQL injection and Session hijacking: SQL injection for website hacking, session hijacking.	2
7.	Using Cryptool to encrypt and decrypt password, implement encryption and decryption using Ceaser Cipher	2
8.	Pen testing: Penetration Testing using Metasploit and metasploitable.	2

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Text Books:

1. EC-Council "Ethical Hacking and Countermeasures Attack Phases", Cengage Learning
2. James S. Tiller, "The Ethical Hack: A Framework for Business Value Penetration Testing", Auerbach Publications, CRC Press.
3. The Basics of Hacking and Penetration Testing, Patrick Engebretson 2nd edition Syngress.
4. Ethical Hacking A Comprehensive Beginner's Guide to Learn and Master Ethical Hacking, By Hilary Morrison, hein smith · 2018, Create Space Independent Publishing Platform

Reference Books:

1. EC-Council, "Ethical Hacking and Countermeasures Attack Phases", Cengage Learning.

2. Michael Simpson, Kent Backman, James Corley, "Hands-On Ethical Hacking and Network Defense", Cengage Learning.

3. Matt Walker, All-In-One-CEH-Certified-Ethical-Hacker-Exam Guide.

Course code: SCS24PCP252 **Course category:** PCC		**Course name:** Operating Systems Lab
Credits: 1	**Teaching scheme:** P-2	**Evaluation scheme:** TW-30, ESE–20
Pre-requisites:		
Proficiency in the C programming language is essential for Linux programming		
Gain a basic understanding of operating system concepts like process.		
Course Objectives:		
The course objective is to analyze fundamentals of the Linux operating system, including installation, configuration, administration, file management, and security.		
Laboratory Outcomes:		
LO1: Install and administer server services		
LO2: Manage Users and Groups;		
LO3: Setup and manage policies		
LO4: Implement File Services		

Sr.No.	Name of Experiment	Practical Hours
1	Download Install and configure Ubuntu LINUX Operating system on your machine.	2
2	Execute General purpose and process related command on your machine	2
3	Execute File and Directory manipulation related command on your machine	2
4	Write a shell script to find greatest among three numbers ,Fibonacci series using conditional statement.	2
5	Write a shell script to list all files in a directory.	2
6	Write a shell script to find whether a give files present. Write a shell script to check read, write permission of given file.	2
7	Write a C Program to calculate total waiting and Turnaround time of n process with FCFS algorithm.	2
8	Simulate algorithm for deadlock prevention	2
9	Simulate a page replacement algorithm.	2
10	Write a C Program to implement FIFO page replacement algorithm	2

Text Books: 1. Abraham Silberschatz, Peter B. Galvin and Greg Gagne, Operating System Concepts, Wiley Publication, 8th Edition, 2008.
2.William Stallings, Operating System: Internals and Design Principles, Prentice Hall, 8thEdition.
3.Andrew S. Tanenbaum, Modern Operating System, PHI Publication, 4th Edition, 2015.
Reference Books: 1.D. M. Dhamdhere, Systems Programming and Operating Systems, McGraw-Hill,2ndEdition, 1996.
2.Garry Nutt, Operating Systems Concepts, Pearson Publication, 3rd Edition, 2003.
3.Harvey M. Deitel, An Introduction to Operating Systems, Addison-Wesley Publication, 2 ndEdition,1990
4.Tom Adelstein and Bill Lubanovic, Linux System Administration, O'ReillyMedia

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Course code : SCS24PCP253 **Course name:** Object Oriented Programming Using C++ Lab **Course category:** PCC
Credits: 1 **Teaching scheme:** P- 2 **Evaluation scheme:** TW-30, **PR**–20
Pre-requisites: Basics of programming concept , Basics of DS, Fundamental of Procedure Oriented
Course Objectives: To impart students with knowledge on basics of programming and Object Oriented Programming concepts.
Laboratory Outcomes: At the end of the course, the students will be able to -
LO1: Implement and apply fundamentals of object oriented programming.
LO2: Implement classes and object concept with real world entity
LO3: Demonstrate concept of inheritance & overloading
LO4: Device simple exceptions handling

Contents –

Sr. No.	List of Experiments	Practical Hours
1	Write a program to demonstrate basic syntactical constructs of C++. ● Operators & Expressions ● Looping Statement ● Decision Making Statement	2
2	Write a program to demonstrate decision making statements and control structure, Arrays in C++.	2
3	Write a program to demonstrate Classes and Objects, methods constructor in C++.	2
4	Write a program to implement function overloading.	
5	Write a program to implement operator overloading.	
6	Write a program to friend function.	2
7	Write a program to implement all types of constructors and destructor.	
8	Write a program to implement inheritance.	2
9	Write a program to implement Polymorphism.	2
10	Write a program to implement exception handling.	2
11	Develop Mini Project	

Text Books: E. Balagurusamy , Object Oriented Programming with C++ , MC Graw Hill Publication
Reference Books: 1. Tony Gaddis, Starting out with C++: from control structures through objects (7e)

2. B. Stroupstrup, The C++ Programming Language, Addison Wesley, 2004.
3. Problem Solving with C++ by Walter Savitch, Addison Wesley
Online Resources: NPTEL/SWAYAM

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Course code: CSB24PCC452	**Course name:** Blockchain	**Course category:** PCC
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Duration of External Exam: 2 Hrs		
Pre-requisites: Computer Network, Cryptography		
Course Objectives:		
1. To understand the fundamentals of blockchain, its evolution, and the concept of decentralization.		
2. To study blockchain architecture, versions, and the concept of smart decentralized ecosystems with cryptocurrencies.		
3. To explore blockchain mechanisms like proof systems, hashing, and Bitcoin's transaction and payment structures.		
4. To compare and design blockchain platforms and assess their applicability in different domains.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Understand the basic structure, history, and types of blockchain systems and decentralization methods.		
CO2: Analyze blockchain architecture, smart contracts, and cryptocurrencies in various real-world use cases.		
CO3: Apply cryptographic principles and consensus algorithms in Bitcoin transactions and wallet mechanisms.		
CO4: Evaluate and design distributed applications using modern blockchain platforms.		

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Contents –

Unit	Content	Teaching hours
1	**Introduction to Blockchain Technology** Discover Blockchain Technology: Blockchain, Growth of blockchain technology, Distributed systems, History of blockchain and Bitcoin, Types of blockchain. Decentralization: Methods of decentralization, Routes of decentralization, Blockchain and full ecosystem decentralization, Smart contracts, Decentralized organizations and platforms for decentralization	8
2	**Decentralization and Smart Ecosystem** Blockchain: Architecture, Versions, Variants, Use cases, Life use cases of blockchain, Blockchain vs shared Database, Introduction to cryptocurrencies, Types, Applications. Bitcoins: Introducing Bitcoin, Bitcoin digital keys and addresses, Transactions, Blockchain mining. Alternative Coins. Limitations of Bitcoin.	7
3	**Blockchain Architecture, Cryptocurrencies, and Bitcoin** Concept of Double Spending, Hashing, Proof of work. Proof of Stake. Bitcoin Network and payments, Bitcoin network, Wallets, Bitcoin payments, Innovation in Bitcoin, Bitcoin Clients and APIs.	8

4	**Blockchain Platforms and Applications** Introduction to Blockchain Platforms: Ethereum, Hyperledger, IOTA, EOS, Multichain, Bigchain, etc. Advantages and Disadvantages, EthereumvsBitcoin, Design a new blockchain, Potential for disruption, Design a distributed application, Blockchain applications.	7

Text Books:

1) I. Bashir, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, 4th ed., Packt Publishing, 2023.

2) A. D. Lewis, Blockchain and the Law: The Rule of Code, Harvard University Press, 2021

Reference Books:

1) M. Swan, Blockchain: Blueprint for a New Economy, O'Reilly Media, 2015.

2) R. Wattenhofer, The Science of the Blockchain, Inverted Forest Publishing, 2016.

3) D. Drescher, Blockchain Basics: A Non-Technical Introduction in 25 Steps, Apress, 2017.

Online Resources:

1. NPTEL / SWAYAM lectures.

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Course code: SCS24VSL255	Course Name: Blockchain
Course Category: VSEC	
Credits: 1 Teaching **scheme:** L-1 P-0 T-0 **Evaluation scheme:** CA–30, ESE–20, TW–0, PR–0	
Pre-requisites: Computer Networks, Cryptography	
Course Objectives:	
1. To understand the fundamentals of blockchain, its evolution, and the concept of decentralization. 2. To study blockchain architecture, versions, and the concept of smart decentralized ecosystems with cryptocurrencies. 3. To explore blockchain mechanisms like proof systems, hashing, and Bitcoin's transaction and payment structures. 4. To compare and design blockchain platforms and assess their applicability in different domains.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand the basic structure, history, and types of blockchain systems and decentralization methods..	
CO2: Analyze blockchain architecture, smart contracts, and cryptocurrencies in various real-world use cases.	
CO3: Apply cryptographic principles and consensus algorithms in Bitcoin transactions and wallet mechanisms.	
CO4: Evaluate and design distributed applications using modern blockchain platforms.	

Contents –

Unit	Content	Teaching hours
1.	**Unit I: Introduction to Blockchain Technology** Discover Blockchain Technology: Blockchain, Growth of blockchain technology, Distributed systems, History of blockchain and Bitcoin, Types of blockchain. Decentralization: Methods of decentralization, Routes of decentralization, Blockchain and full ecosystem decentralization, Smart contracts, Decentralized organizations and platforms for decentralization	4
2.	**Unit II: Decentralization and Smart Ecosystem** Blockchain: Architecture, Versions, Variants, Use cases, Life use cases of blockchain, Blockchain vs shared Database, Introduction to cryptocurrencies, Types, Applications. Bitcoins: Introducing Bitcoin, Bitcoin digital keys and addresses, Transactions, Blockchain mining. Alternative Coins. Limitations of Bitcoin.	3

3.	**Unit III: Blockchain Architecture, Cryptocurrencies, and Bitcoin** Concept of Double Spending, Hashing, Proof of work. Proof of Stake. Bitcoin Network and payments, Bitcoin network, Wallets, Bitcoin payments, Innovation in Bitcoin, Bitcoin Clients and APIs.	4
4.	**Unit IV: Blockchain Platforms and Applications** Introduction to Blockchain Platforms: Ethereum, Hyperledger, IOTA, EOS, Multichain, Bigchain, etc. Advantages and Disadvantages, EthereumvsBitcoin, Design a new blockchain, Potential for disruption, Design a distributed application, Blockchain applications..	3

Text Books:

1) I. Bashir, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, 4th ed., Packt Publishing, 2023.

2) A. D. Lewis, Blockchain and the Law: The Rule of Code, Harvard University Press, 2021

Reference Books:

1) M. Swan, Blockchain: Blueprint for a New Economy, O'Reilly Media, 2015.

2) R. Wattenhofer, The Science of the Blockchain, Inverted Forest Publishing, 2016.

3) D. Drescher, Blockchain Basics: A Non-Technical Introduction in 25 Steps, Apress, 2017.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: SCS24VSP255	**Course Name: Blockchain Lab**
Course category: Vocational Skill and Skill Enhancement course	
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation scheme:**CA–0, MSE–0, ESE–0, TW–30, PR–20	
Pre-requisites: Computer Networks, Cryptography	

Course Objectives:

1. To understand the fundamentals of blockchain, its evolution, and the concept of decentralization.
2. To study blockchain architecture, versions, and the concept of smart decentralized ecosystems with cryptocurrencies.
3. To explore blockchain mechanisms like proof systems, hashing, and Bitcoin's transaction and payment structures.
4. To compare and design blockchain platforms and assess their applicability in different domains.

Laboratory Outcomes: At the end of the session, the students will be able to -

CO1: Understand the basic structure, history, and types of blockchain systems and decentralization methods..

CO2: Analyze blockchain architecture, smart contracts, and cryptocurrencies in various real-world use cases.

CO3: Apply cryptographic principles and consensus algorithms in Bitcoin transactions and wallet mechanisms.

CO4: Evaluate and design distributed applications using modern blockchain platforms

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	To install and configure blockchain development tools.	2
2.	To understand blockchain structure and block linking using Python.	2
3.	To simulate decentralized nodes communicating in a blockchain network.	2
4.	To deploy a simple smart contract on Ethereum using Remix IDE.	2
5.	To perform ether transactions between accounts.	2
6.	To simulate a Bitcoin-like transaction and mining process.	2

7.	To demonstrate consensus algorithms.	2
8.	To create a customized blockchain prototype	2
9.	To develop and deploy a simple DApp using Solidity and web3.js.	2
10.	To explore and compare platforms like Ethereum, Hyperledger, and IOTA.	2

Text Books:

1) I. Bashir, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, 4th ed., Packt Publishing, 2023.

2) A. D. Lewis, Blockchain and the Law: The Rule of Code, Harvard University Press, 2021

Reference Books:

1) M. Swan, Blockchain: Blueprint for a New Economy, O'Reilly Media, 2015.

2) R. Wattenhofer, The Science of the Blockchain, Inverted Forest Publishing, 2016.

3) D. Drescher, Blockchain Basics: A Non-Technical Introduction in 25 Steps, Apress, 2017.

Online Resources: 1.NPTEL / SWAYAM lectures.

Exit 1 Syllabus

Course code: SCS4VSI101 **Course name:** Computer Hardware and Maintenance **Course category:** VSC
Credits: 8 **Teaching scheme:** P-16 hrs **Evaluation scheme:** TW–60, PR–40
Pre-requisites: Basic Introduction of Computer.
Course Objectives:
1. To Provide an in-depth understanding of computer hardware and maintenance of peripherals.
2. To understand the functionality of computer software / hardware and troubleshooting while using it.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Learn key concepts, technologies of computer hardware and do the maintenance of it.
CO2: Troubleshoot the issues in system software and resolve it.
CO3: Use various tools and methods to assemble and disassemble hardware components of computer.

Unit	Content
1	Module 1: Introduction to Hardware Components Overview of computer hardware components (CPU, RAM, motherboard, etc.) , Understanding peripheral devices (keyboard, mouse, monitor, etc.), Introduction to hardware terminology and concepts
2	Module 2: Hardware Assembly Techniques for safely assembling and disassembling computer hardware, Connecting peripheral devices and cables.
3	Module 3: Hardware Installation (Drivers / Firmware) Installing and configuring CPU, RAM, motherboard, and power supply, Basic troubleshooting during assembly
4	Module 4: Hardware Troubleshooting Identifying common hardware issues (e.g., overheating, hardware conflicts), Using diagnostic tools (e.g. BIOS diagnostics)
5	Module 5: Hardware Diagnostics Troubleshooting techniques for hardware problems , Repair and replacement of faulty hardware components

6	Module 6: Operating System Installation and Configuration (Windows) Installing an operating system, Understanding partitioning and formatting of hard drives Device drivers installation and updates , Basic operating system configurations for optimal hardware performance
7	Module 7: Operating System Installation and Configuration (Open Source,(Linux /ubuntu etc.) Installing an operating system, Understanding partitioning and formatting of hard drives , Device drivers installation and updates ,Basic operating system configurations for optimal hardware performance.
8	Module 8: Installing Software and Peripherals Installing Printers /Scanners etc. , Common system / application softwares installation and un installation. Driver updates etc. OS updates.
9	Module 9: User Management and Privacy Manage the resources of a Computer System (Windows and or Linux) according to users need. Security Settings etc.
10	Module 10: Preventive Maintenance and Upgrades Importance of preventive maintenance for hardware longevity, Cleaning and dusting hardware components, Scheduled maintenance tasks (e.g., updating firmware, replacing CMOS battery) Planning and executing hardware upgrades (e.g., RAM, storage, graphics card)
11	Module 11: Data Backup and Recovery Importance of data backup , Implementing backup strategies (local backups, cloud backups) Data recovery techniques in case of hardware failure or data loss, Using backup and recovery software effectively
12	Module 12: OS security Measures Windows Firewall Configurations and Antivirus Installation and Settings , Identify different types of Virus attack and take suitable course of action to disinfect a system.
13	Module 13: Introduction of Transmission Media and Sharing of resources. Connect computers in a networking system and able to share, files, printers or other resources in a system.
14	Module 14: Network Troubleshooting Troubleshooting network connectivity issues, Basic network security practices.

15	Module 15: Networking and Connectivity Understanding network hardware (cables, Connectors, Tools), Setting up wired networks
16	Module 16: Wireless Devices and Network Management Understanding network hardware (routers, switches, modems), and wireless network Setup.

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Exit 2 Syllabus

Course code: SCS24VSI201 **Course name:** Computer Network & Cyber Security **Course category:** VSC	
Credits: 8 **Teaching scheme:** P-16 hrs/week **Evaluation scheme:** TW–60, PR–40	
Pre-requisites: Fundamentals of Computer Networking.	
Course Objectives:	
1.	Provides an in-depth understanding of computer networks and cyber security principles
2.	To understand the security components and apply them when evaluating a given security mechanism
Course Outcomes: At the end of the course, the students will be able to	
CO1: Learn key concepts, technologies, and strategies to secure information systems	
CO2: Develop basic understanding of security, cryptography, system attacks and defenses against them	
CO3: Identify the security issues in the network and resolve it.	

Contents –

Unit	Content
1	**Introduction to Computer Networks** : Overview of computer, networks Types of networks (LAN, WAN, MAN), Network topologies and protocols, OSI and TCP/IP models, Network devices and their functions
2	**Network Design and Implementation:** Network design principles, Subnetting and addressing, Routing and switching, Wireless networks, Network troubleshooting and tools
3	**Internet and Web Technologies:** Internet architecture and services, HTTP, HTTPS, DNS, Web application architecture, Security considerations in web development
4	**Internet- Security Protocols:** Introduction, Secure Socket Layer(SSL),Transport Layer Security(TLS),Secure Hyper Text Transfer Protocol(SHTTP),Secure Electronic Transaction(SET), SSLversus SET,3-D Secure Protocol, Email Security, WAP Security, Security in GSM, Security in 3G, IEEE 802.11 Security.
5	**Network Security Fundamentals:** Introduction to network security, Authentication and access control, Encryption and cryptography, Firewalls and intrusion detection systems, Virtual Private Networks (VPNs), Intrusion, Case Study: IP Spoofing attacks.
6	**Introduction to Cyber security:** Overview of cyber security, Cyber security landscape and challenges, Importance of cyber security in modern society, Ethical and legal considerations in cyber security.
7	**Cyber security Threats and Attacks:** Common cyber security, threats phishing, ransom ware, Malware, viruses, and worms, Social engineering attacks, Denial of Service (DoS) attacks, Cyber security incident response
8	**Cyber security Policies and Procedures:** Security policies and standards, Risk management, Incident response planning, Legal and ethical considerations in cyber security

9	**Network Monitoring and Management:** Network monitoring tools, Performance optimization, Network management protocols (SNMP), Case studies on network management.

Text Books: 1. Atul Kahate, Cryptography and Network Security, 3e, McGraw Hill Education

2. John W. Rittinghouse, William M.Hancock, "Cyber security Operations Handbook", Elsevier Pub.

3. Roberta Bragg, Mark Rhodes-Ousley, Keith Strassberg , "The Complete reference – Network Security" , Tata McGraw Hill publication

Reference Books: 1. "Network Security Essentials" by William Stallings

2. Behrouz A. Forouzan, Debdeep Mukhopadhyay, Cryptography and Network Security, McGraw Hill Education.

3.Vivek Sood, 'Cyber Law Simplified', McGraw Hill Education.

Online Resources: NPTEL / SWAYAM

Suggestive List of Practical

1	Study of different types of Network cables (Cat5,Cat6 & Fiber optics)and practically implement the cross-wired cable and straight wired cable using crimping tool with Fiber Optics Splicing (joining)
2	Install and configure wired and wireless NIC and transfer files between system LAN and Wireless LAN
3	Install and configure different Network Devices: Hub, Switch, Router etc.
4	Connect the computers in LAN /Configure LAN Network for the LAB.
5	Configure Host IP, Subnet mask and default gateway in a system in LAN(TCP/IP configuration)
6	Establish peer to peer network connection using two systems using switch and router in a LAN.
7	Configure internet connection and use IPCONFIG, PING/Tracer and Net stat utilities to debug the network issues.(also study basic networking commands)
8	Transfer files between systems in LAN using FTP configuration, install print server in a LAN and share the printer in a network.
9	Configure network topology using packet tracer software.
10	Install and use Nmap and Wire shark tool for OS Detection , monitoring network packet and analyze network traffic
11	Install and configure antivirus software, setup operating system update, backup and

	restore, set up password/security for user file/folder or applications.
12	Install and configure firewall settings.
13	Trace the origin of E-mail(tracerpro), website path using different open source tools(tracert utility)
14	Study of various ethical hacking tools
15	Study different cyber forensic tools and Case study on cyber crime
16	Study of cyber security Laws

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Exit 3 Syllabus

Course code: SCS24VSI301 **Course name:** Data Analytics **Course category:** VSC
Credits: 6 **Teaching scheme:** P-12 hrs **Evaluation scheme:** TW–60, PR–40
Pre-requisites: Discrete Mathematics, Good Programming Skill
Course Objectives:
1. To provide students with a foundational understanding of Data Analytics, its importance, and the process of extracting meaningful insights from data.
2. To conduct Exploratory Data Analysis (EDA) & execute a comprehensive data analytics project.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Develop skills to create visually appealing and informative charts, graphs, and dashboards.
CO2: Acquire the ability to receive user inputs and perform essential numerical operations.
CO3: Demonstrate proficiency in performing data pre-processing operations to clean and prepare datasets for analysis.
CO4: Develop the capability to conduct Exploratory Data Analysis (EDA)
CO5: Acquire knowledge of model evaluation techniques to assess the accuracy and reliability of predictive models.
Contents –

Ex. No.	Title
1	Study and implementation of Data Visualization.
2	To get the input from the user and perform numerical operations MAX, MIN,MODE , AVG, SUM, SQRT, ROUND, VARIANCE etc.
3	Reading Dataset or an Excel data sheet and performing data pre-processing operations.
4	Perform data import/export (.CSV, .XLS, .TXT) operations using data frames.
5	Implementation of Linear regression for prediction.
6	Implementation of Logistic regression.
7	Perform K-Means clustering operations and visualize data sets.
8	Build a decision tree model that predicts desired results. Perform model evaluation.
9	**Case Study** - e.g. To identify patterns which indicate if a client has difficulty paying their installments which may be used for taking actions such as denying the loan, reducing the amount of loan, lending (to risky applicants) at a higher interest rate, etc. This will ensure that the consumers capable of repaying the loan are not rejected. Identification of such applicants using EDA is the aim of this case study. In other words, the company wants to understand the driving factors (or driver variables) behind loan default, i.e. the variables which are strong indicators of default.

10	Use a dataset and design CNN Model for classification.
11	Hands on experience of any data analytics tool.
12	Mini Project

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